

BFS Traversal

Step 1: Start at E

Legend

Complete

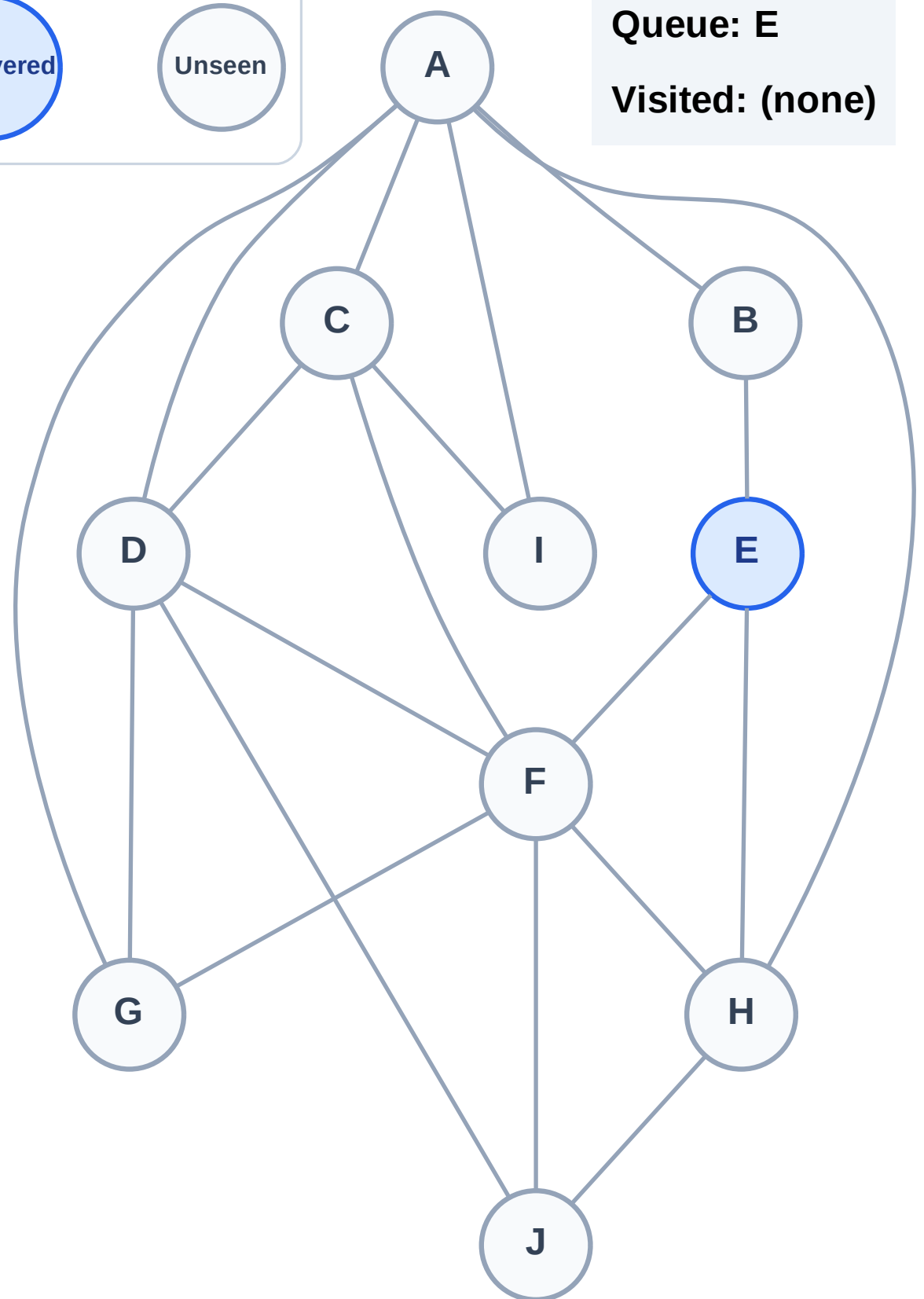
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

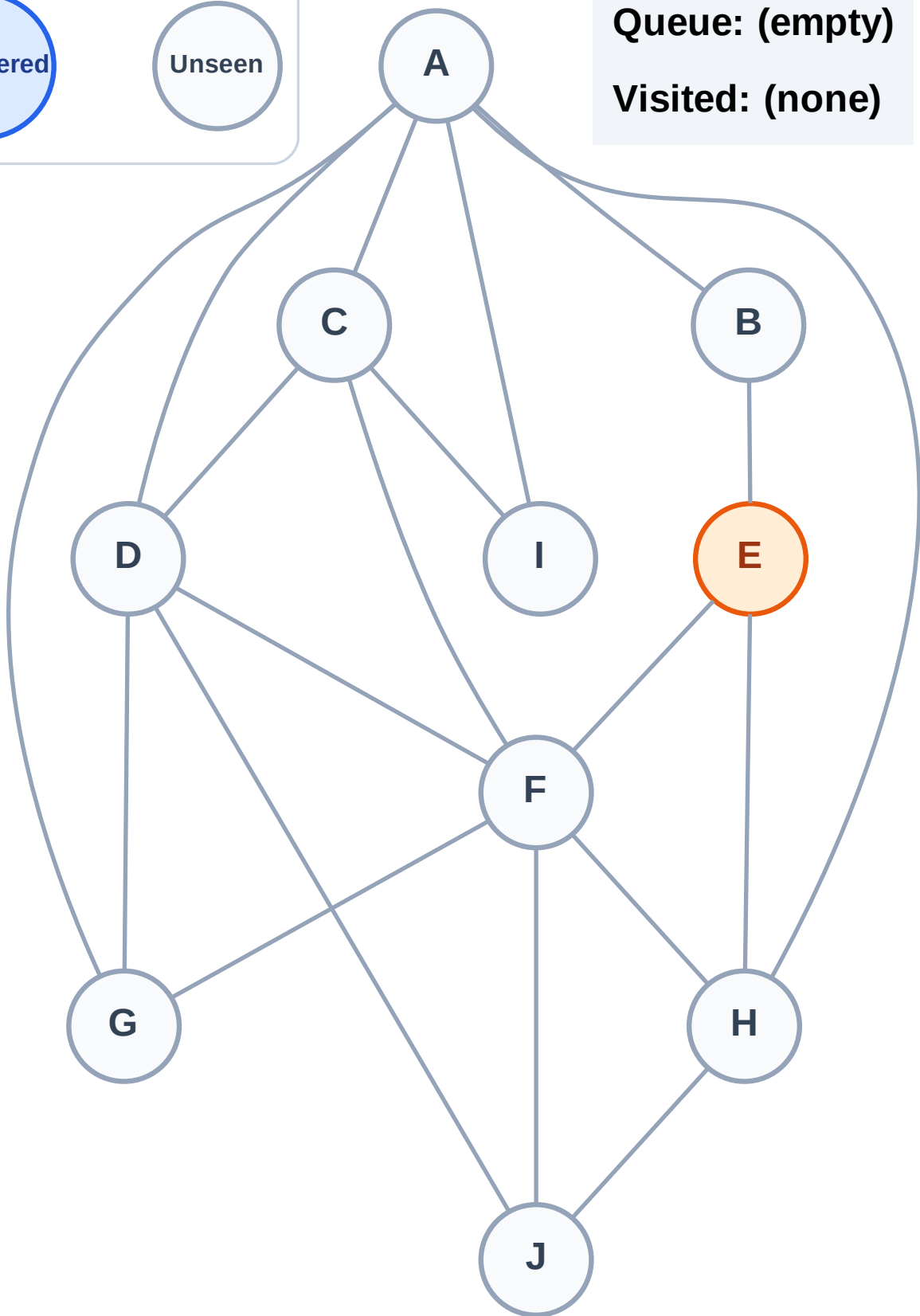
Step 2: Process node E

Legend



Queue: (empty)

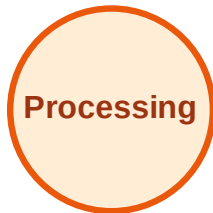
Visited: (none)



BFS Traversal

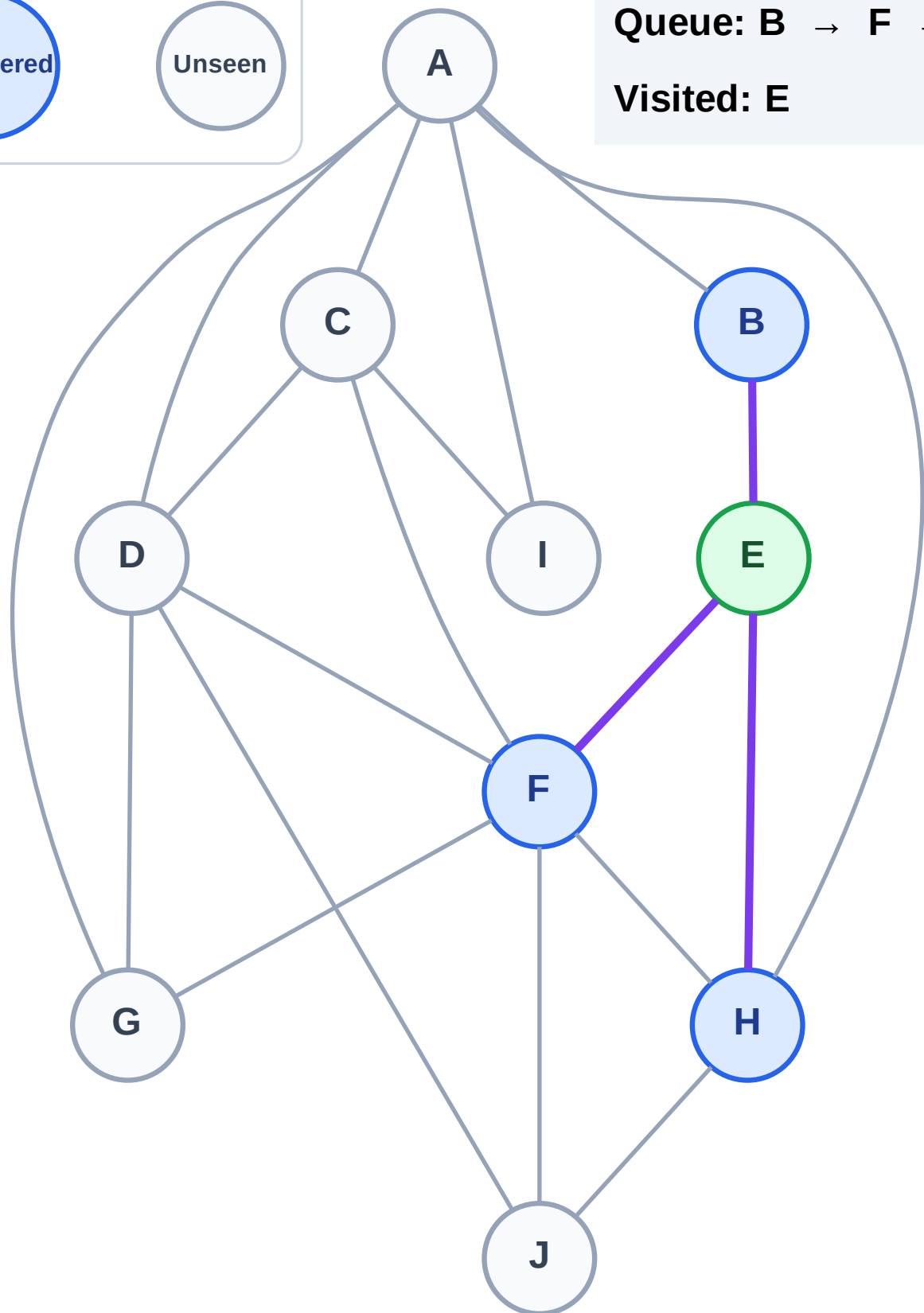
Step 3: Discover B, F, H from E

Legend



Queue: B → F → H

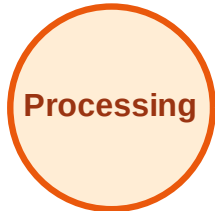
Visited: E



BFS Traversal

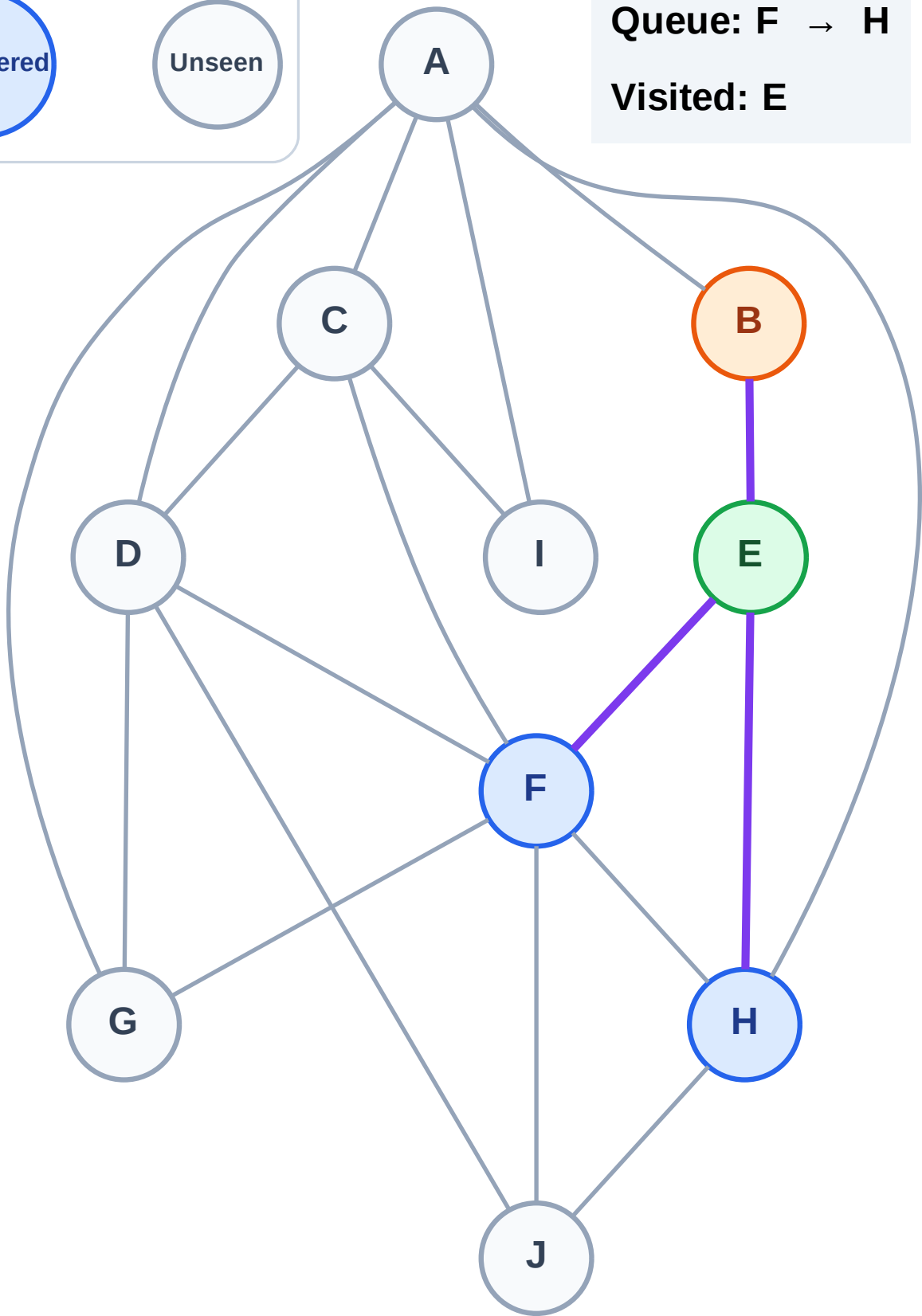
Step 4: Process node B

Legend



Queue: F → H

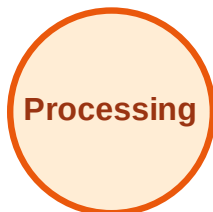
Visited: E



BFS Traversal

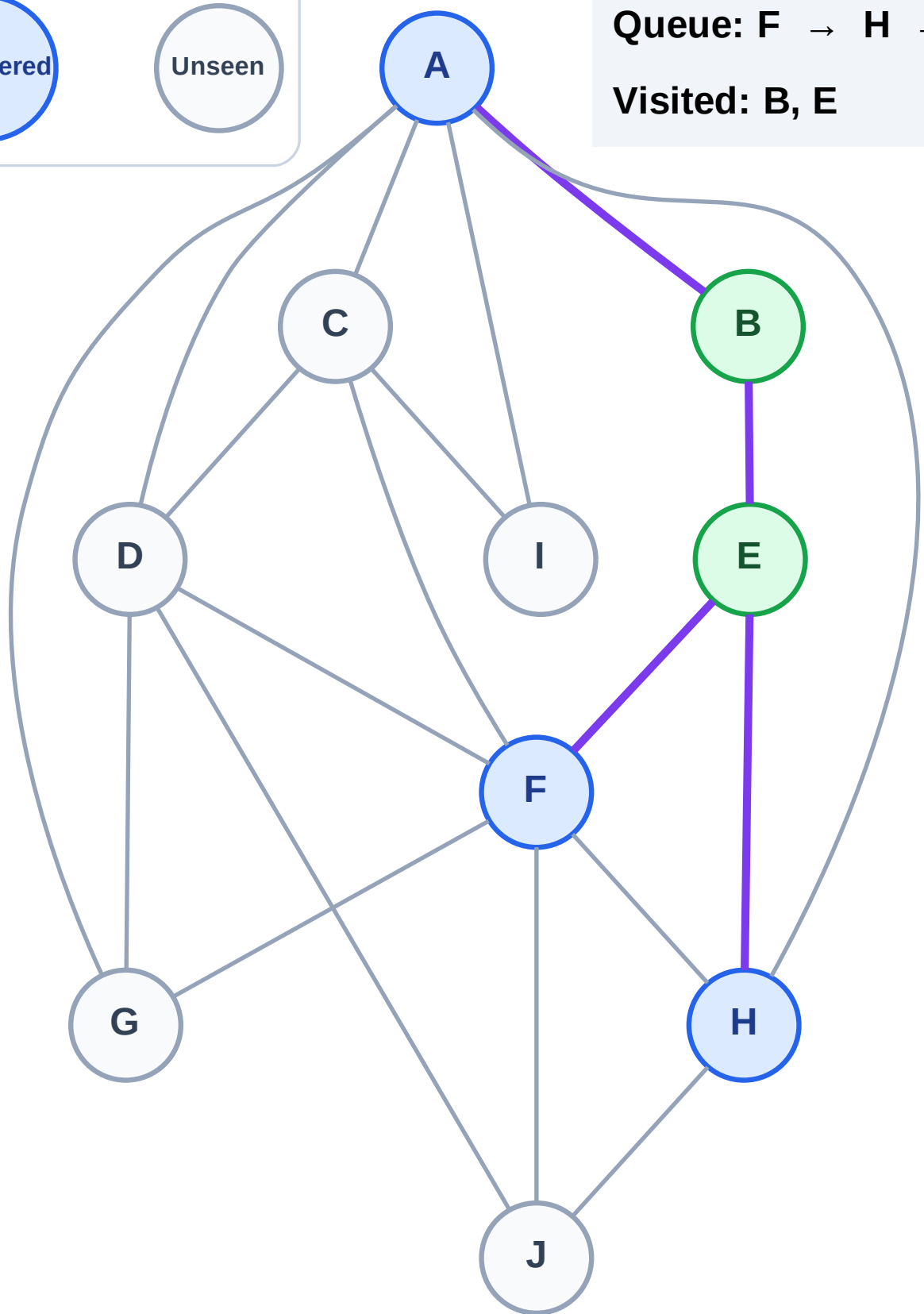
Step 5: Discover A from B

Legend



Queue: F → H → A

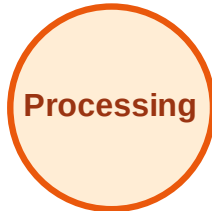
Visited: B, E



BFS Traversal

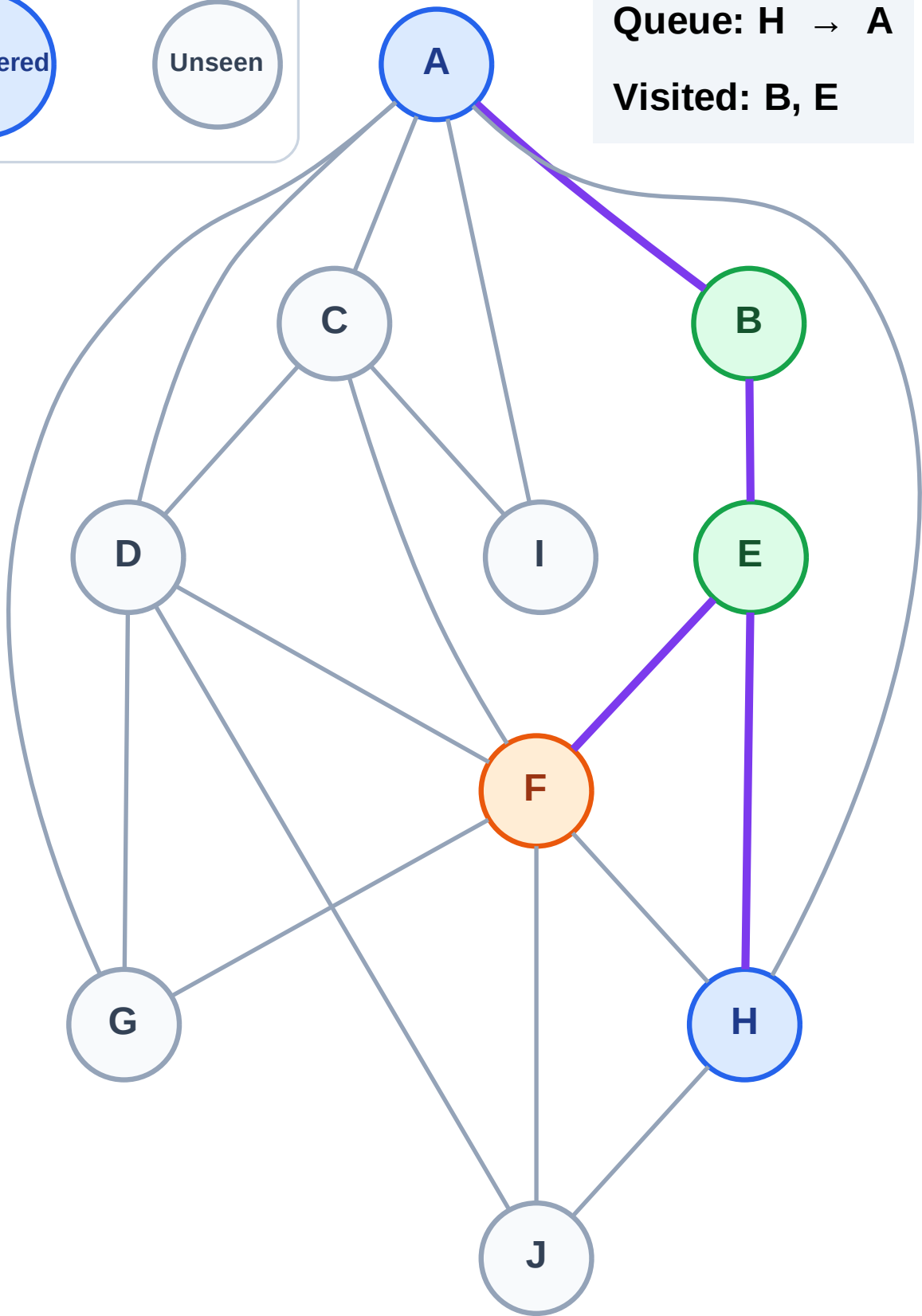
Step 6: Process node F

Legend



Queue: H → A

Visited: B, E



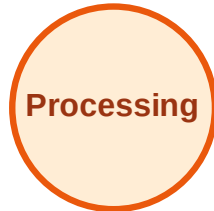
BFS Traversal

Step 7: Discover C, D, G, J from F

Legend



Complete



Processing



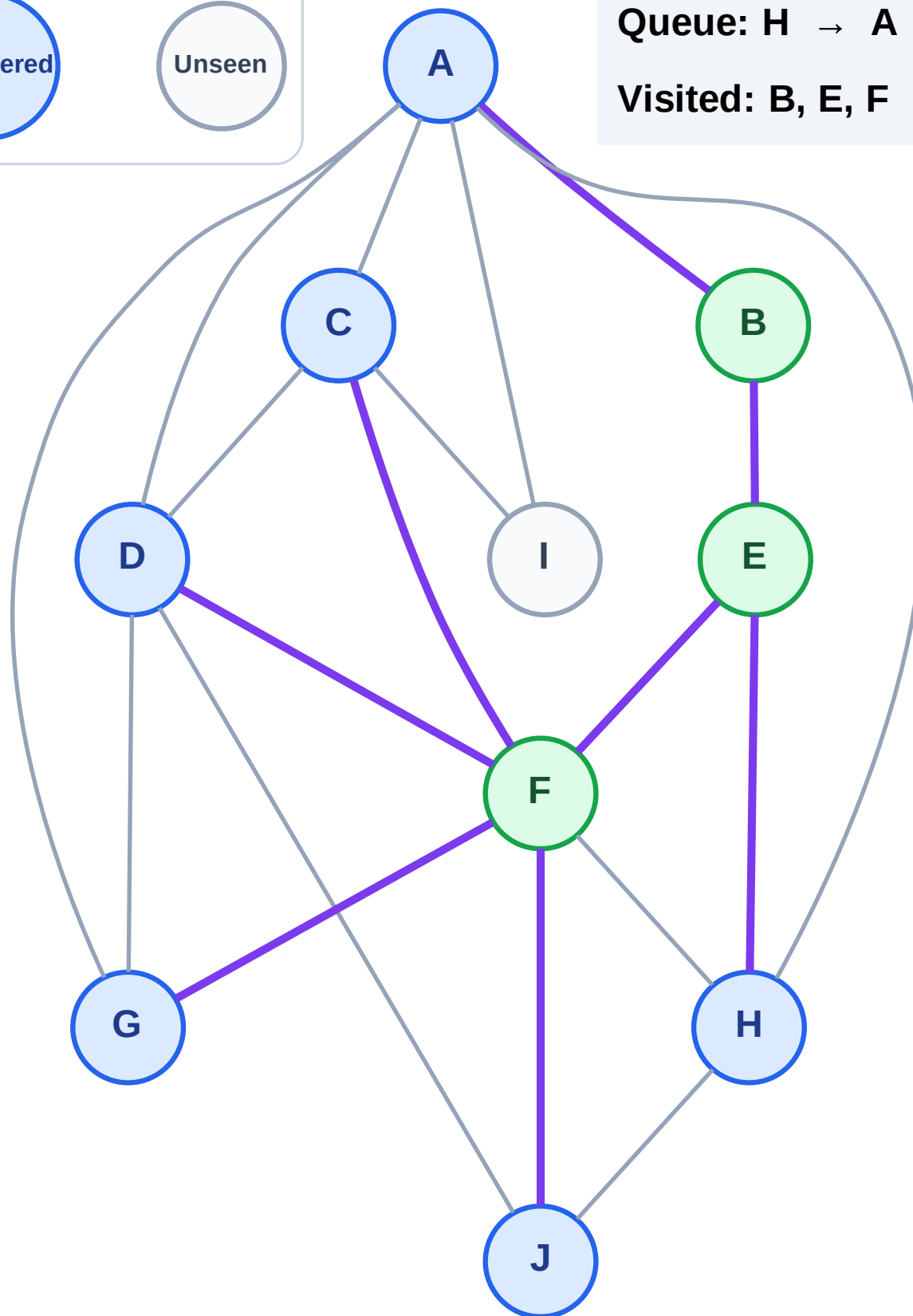
Discovered



Unseen

Queue: H → A → C → D → G → J

Visited: B, E, F



BFS Traversal

Step 8: Process node H

Legend

Complete

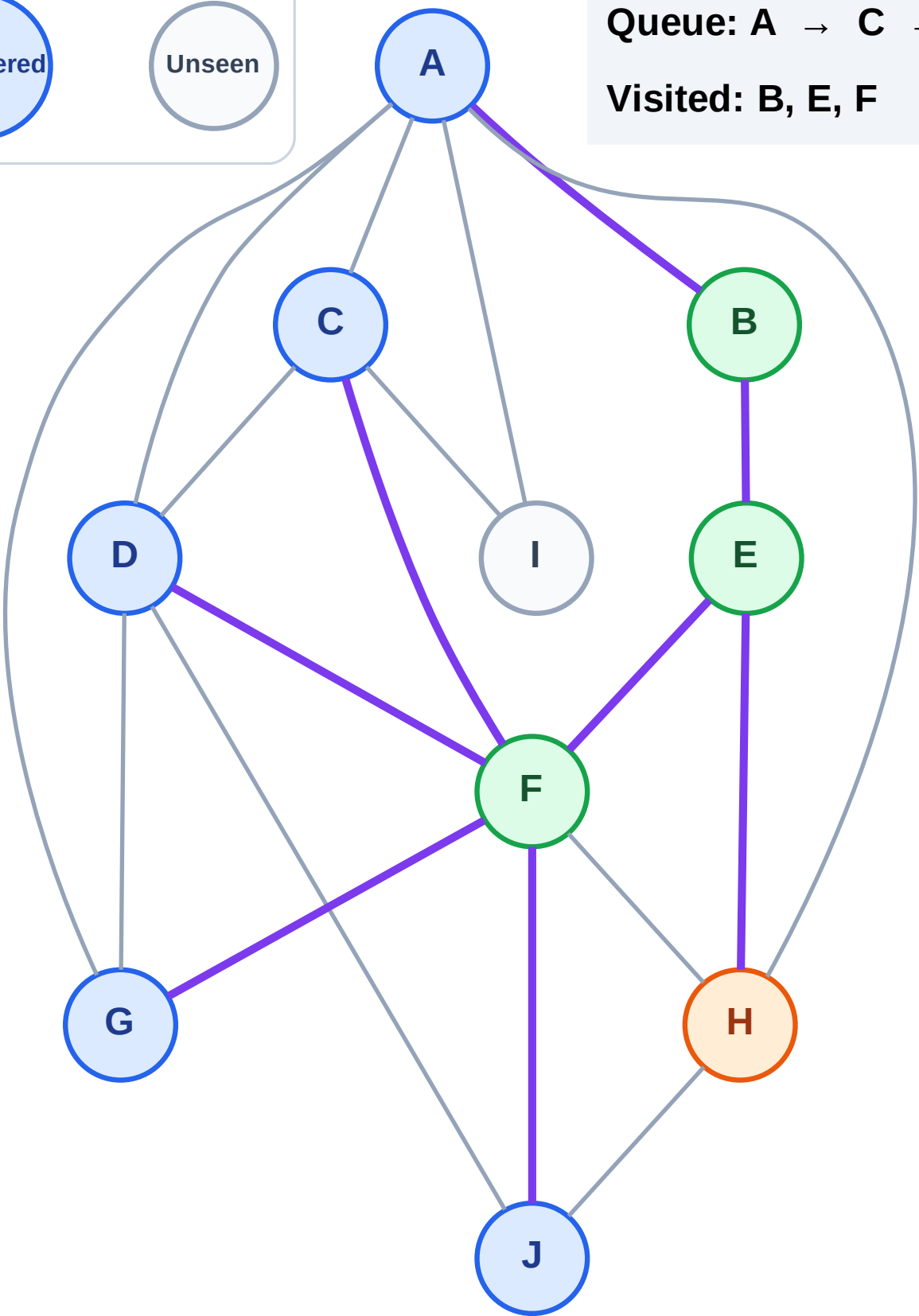
Processing

Discovered

Unseen

Queue: A → C → D → G → J

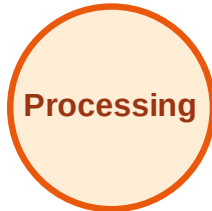
Visited: B, E, F



BFS Traversal

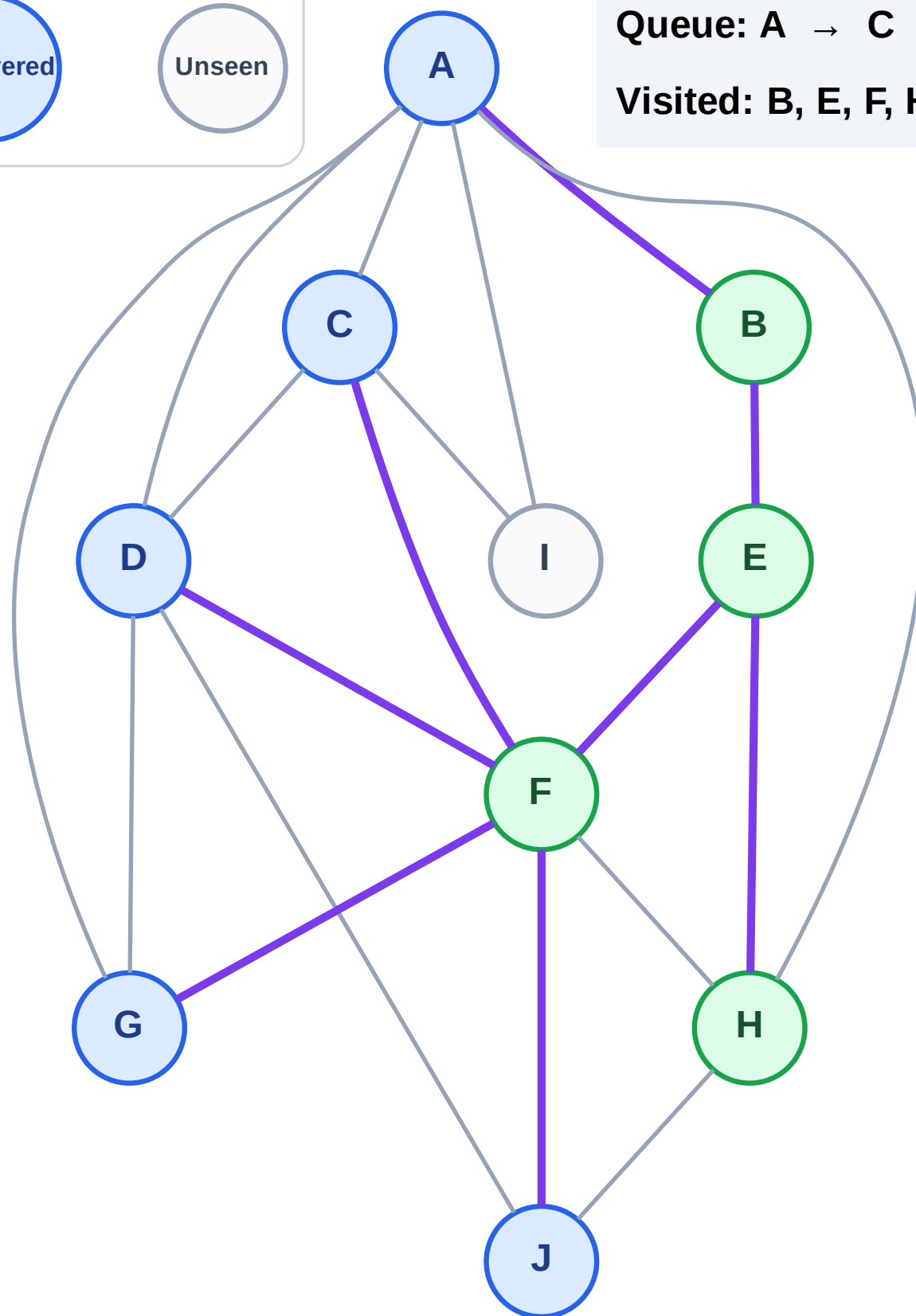
Step 9: No new neighbors from H

Legend



Queue: A → C → D → G → J

Visited: B, E, F, H



BFS Traversal

Step 10: Process node A

Legend

Complete

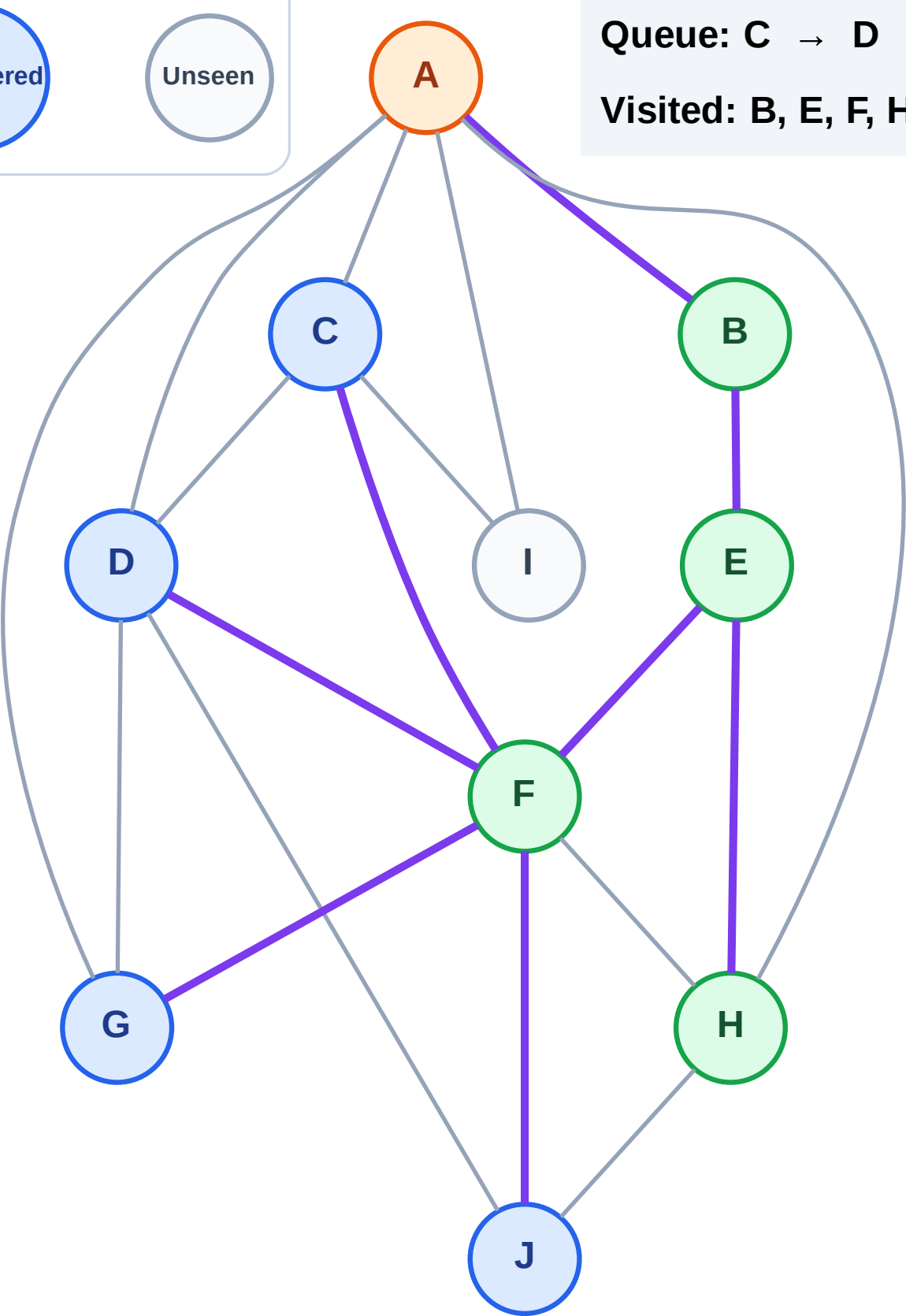
Processing

Discovered

Unseen

Queue: C → D → G → J

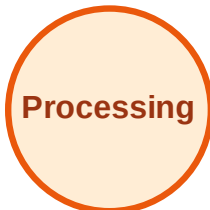
Visited: B, E, F, H



BFS Traversal

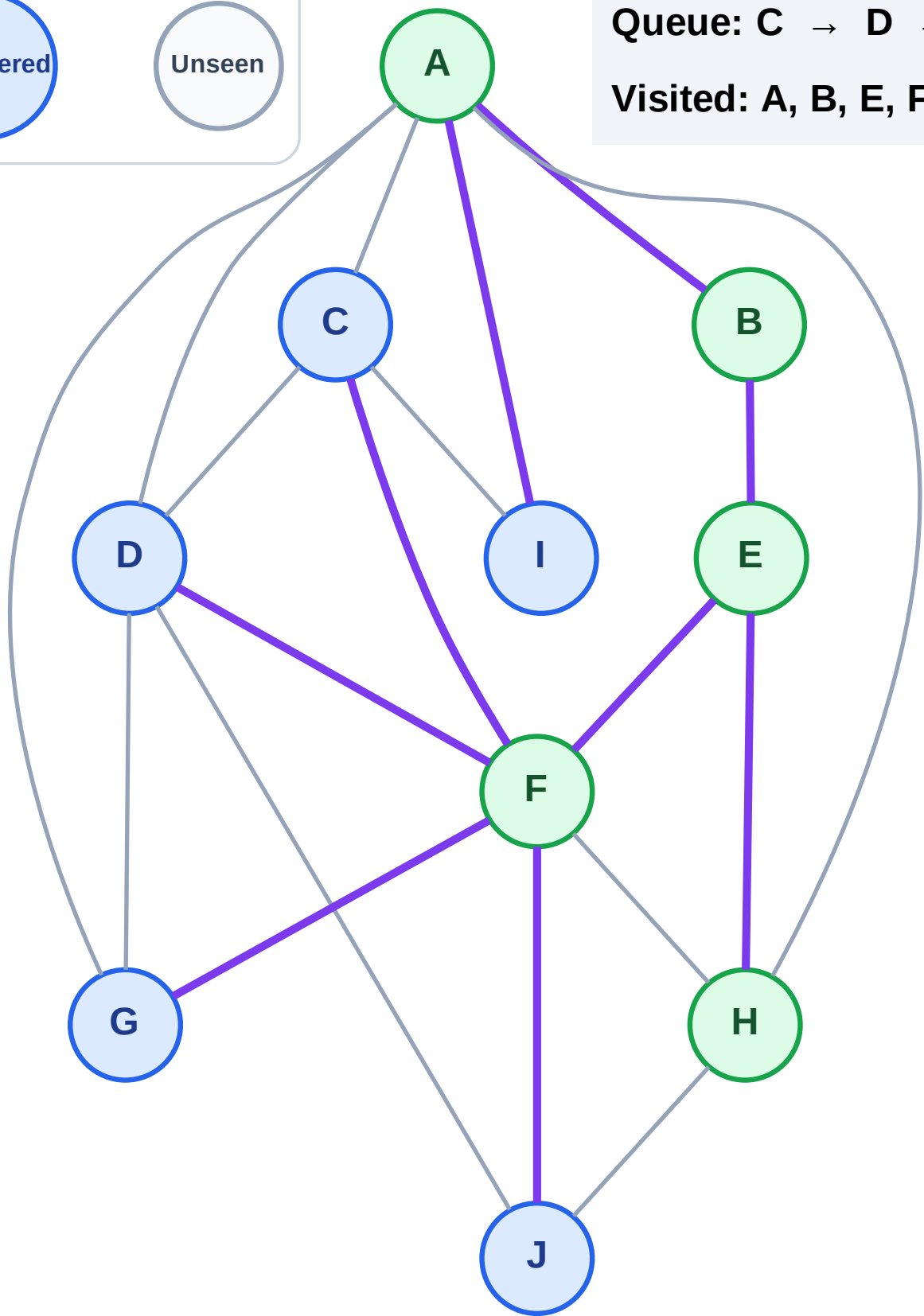
Step 11: Discover I from A

Legend



Queue: C → D → G → J → I

Visited: A, B, E, F, H



BFS Traversal

Step 12: Process node C

Legend

Complete

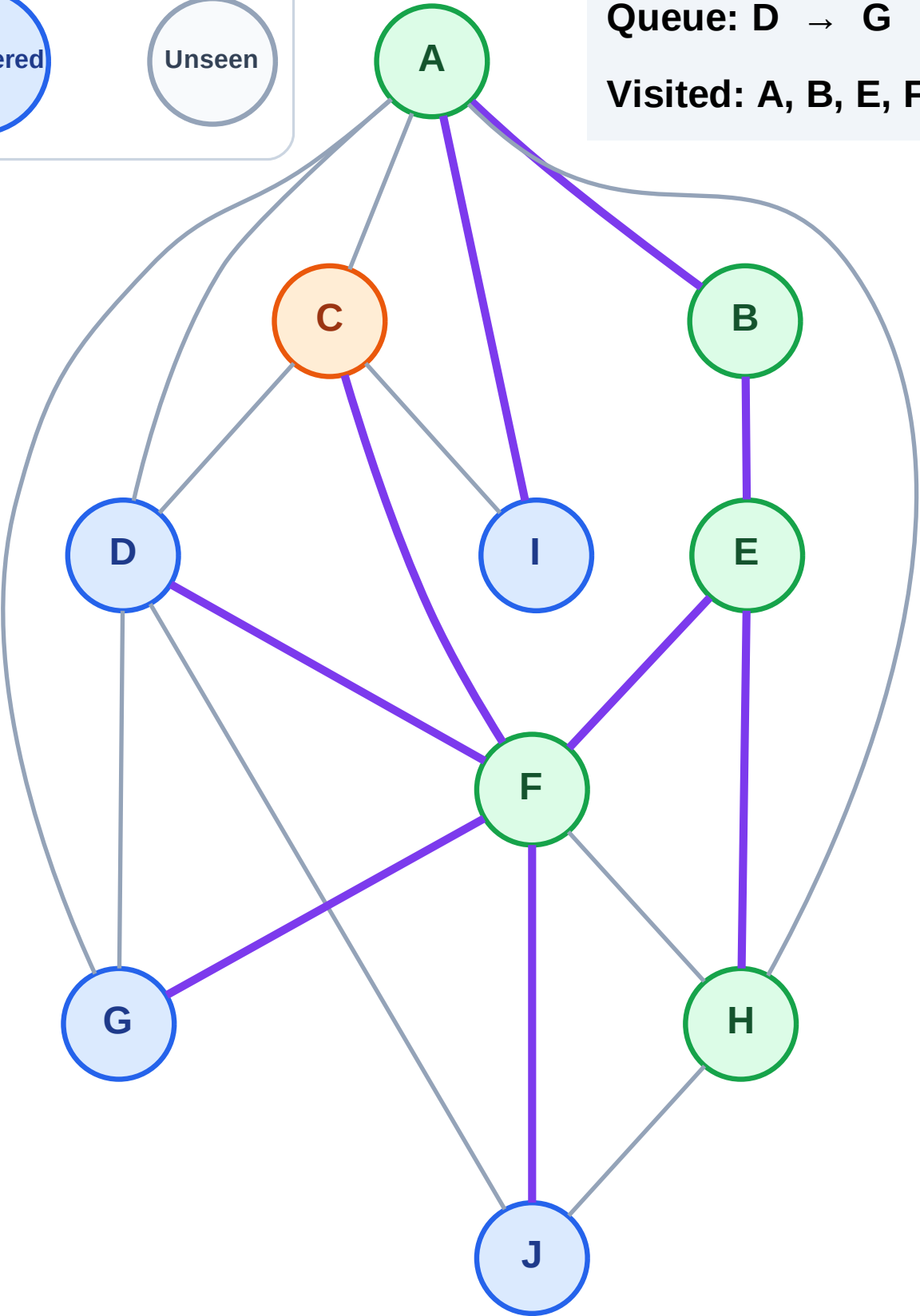
Processing

Discovered

Unseen

Queue: D → G → J → I

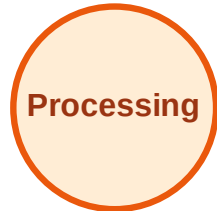
Visited: A, B, E, F, H



BFS Traversal

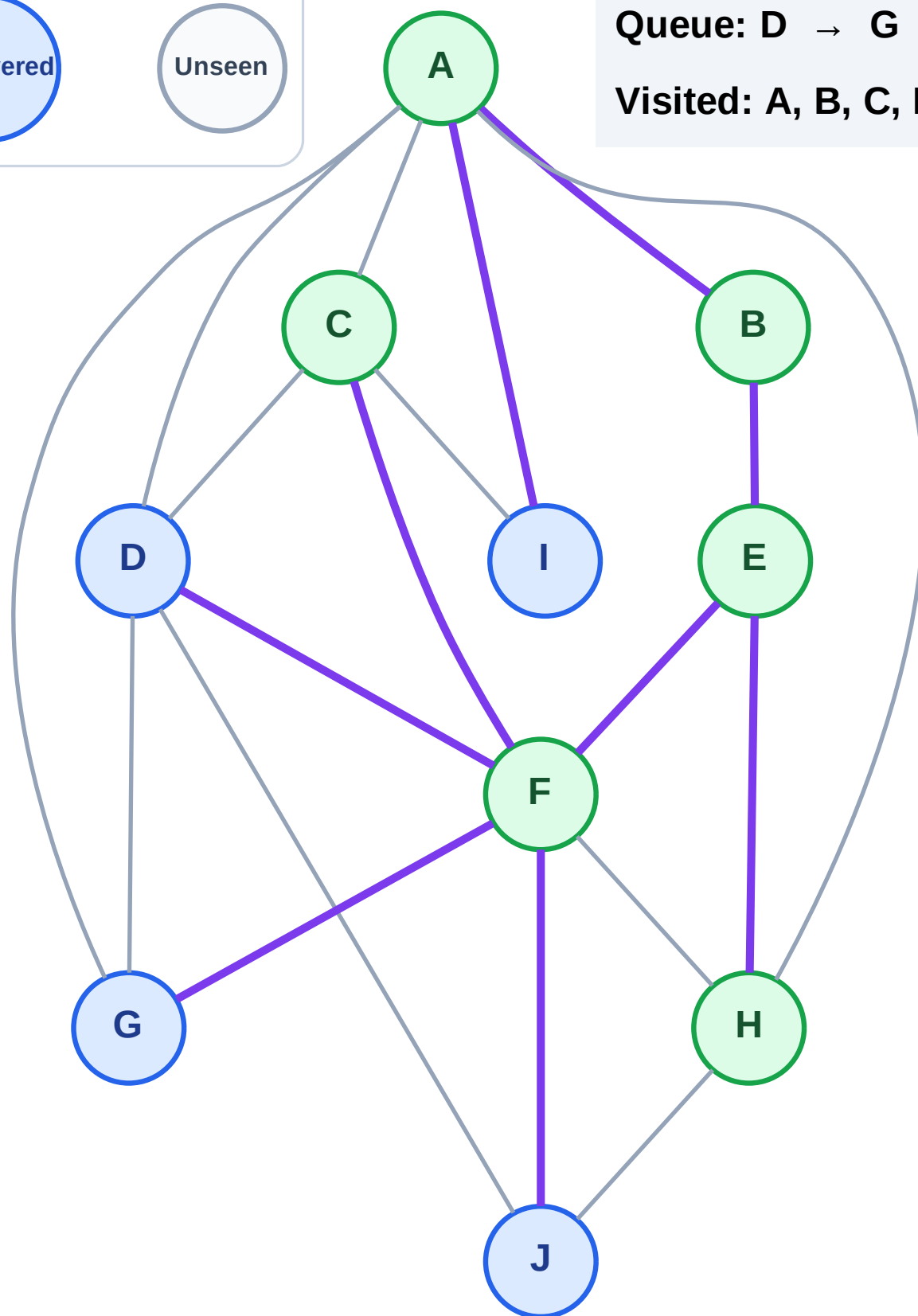
Step 13: No new neighbors from C

Legend



Queue: D → G → J → I

Visited: A, B, C, E, F, H



BFS Traversal

Step 14: Process node D

Legend

Complete

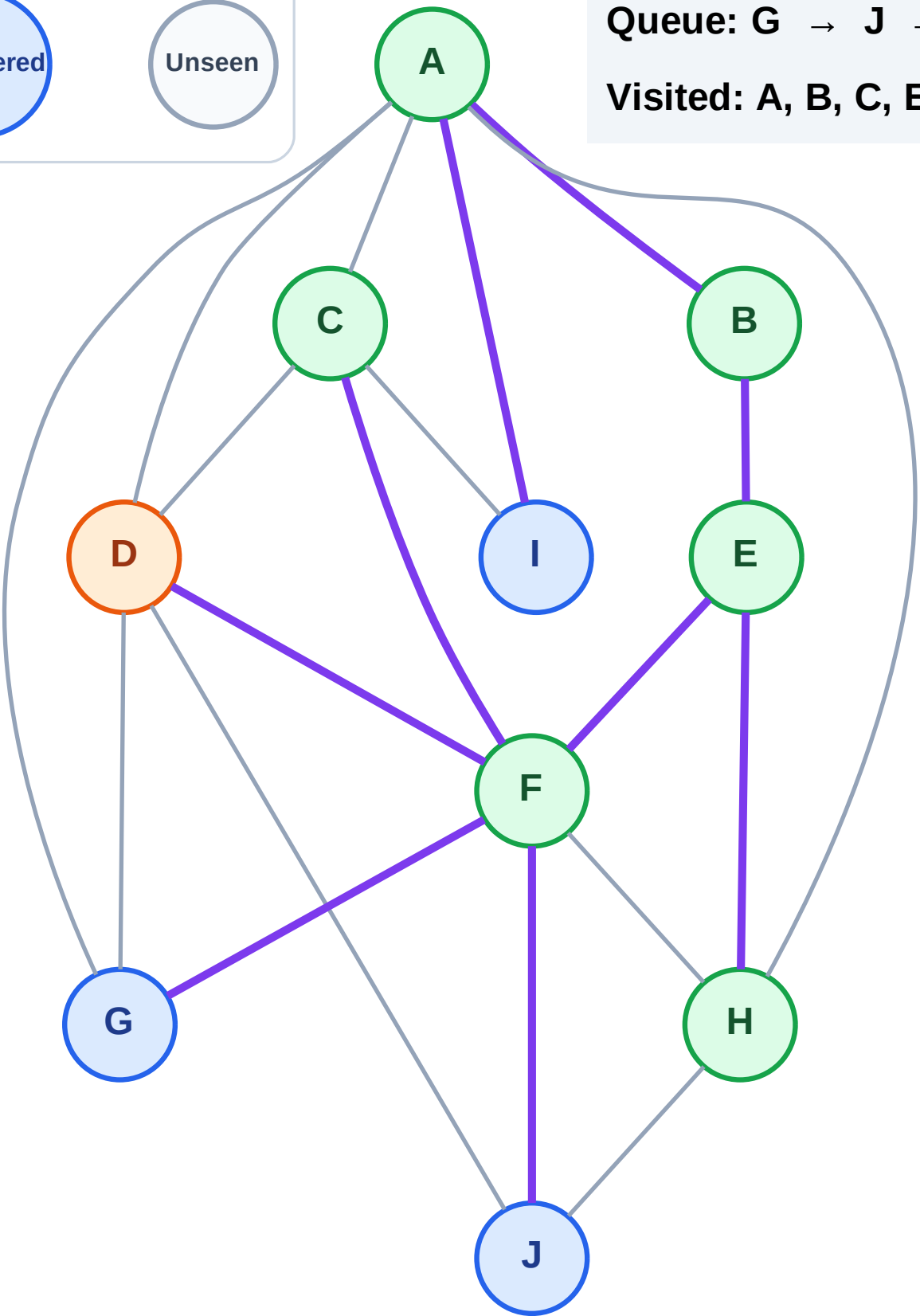
Processing

Discovered

Unseen

Queue: G → J → I

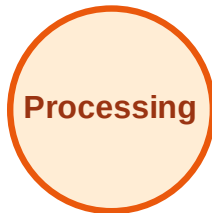
Visited: A, B, C, E, F, H



BFS Traversal

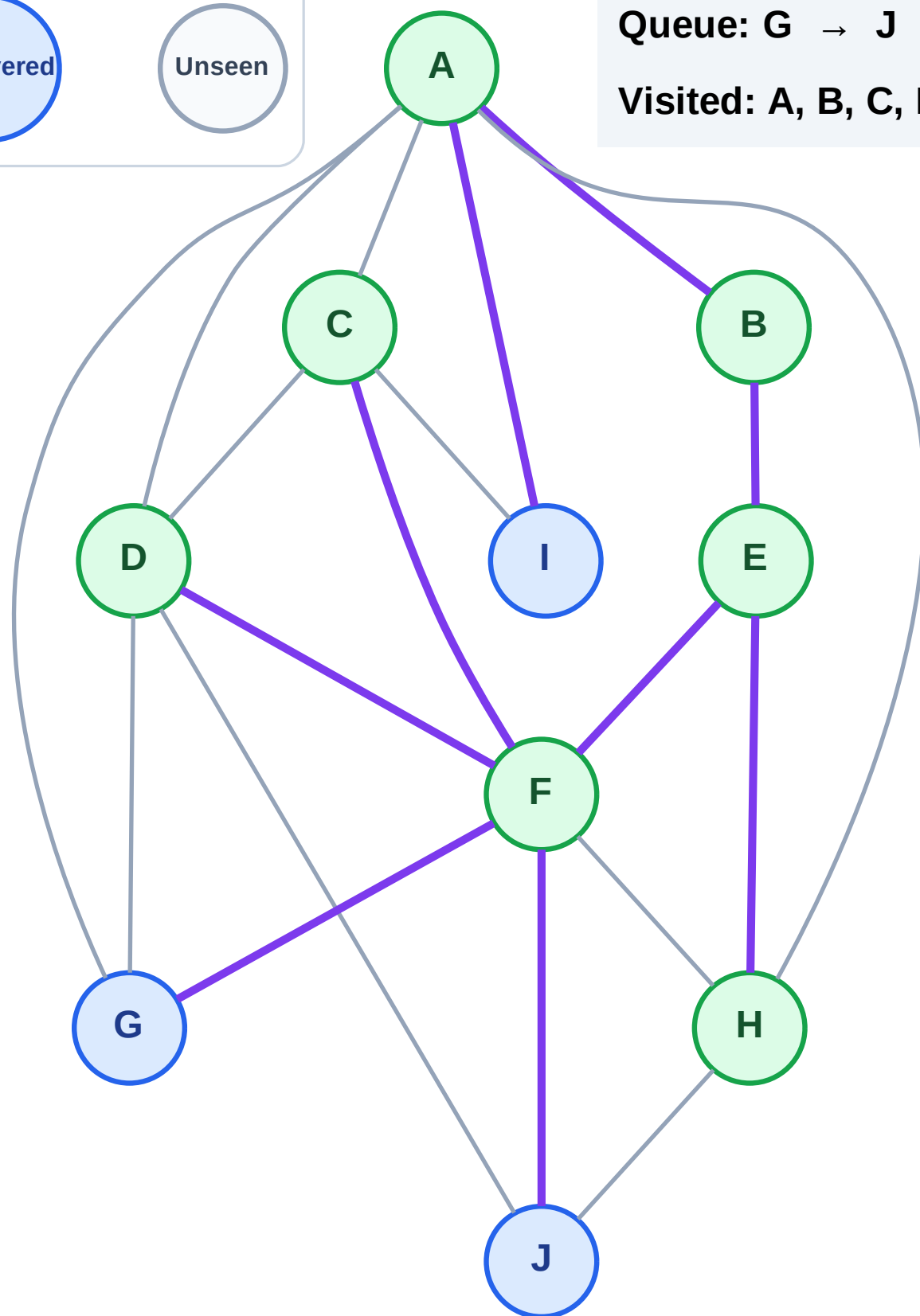
Step 15: No new neighbors from D

Legend



Queue: G → J → I

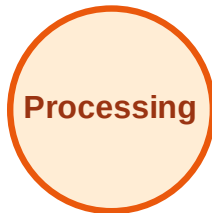
Visited: A, B, C, D, E, F, H



BFS Traversal

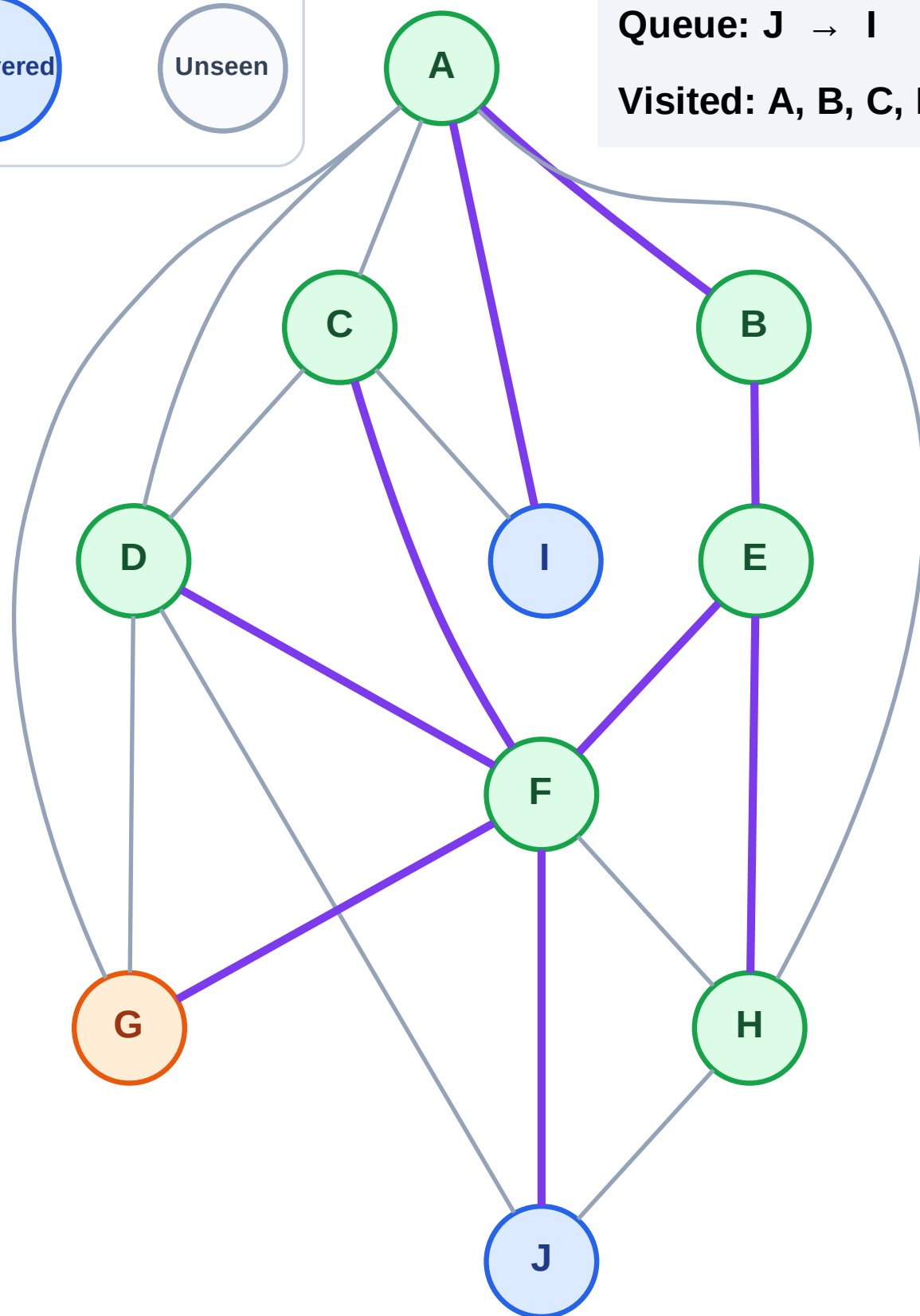
Step 16: Process node G

Legend



Queue: J → I

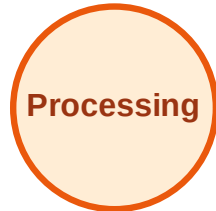
Visited: A, B, C, D, E, F, H



BFS Traversal

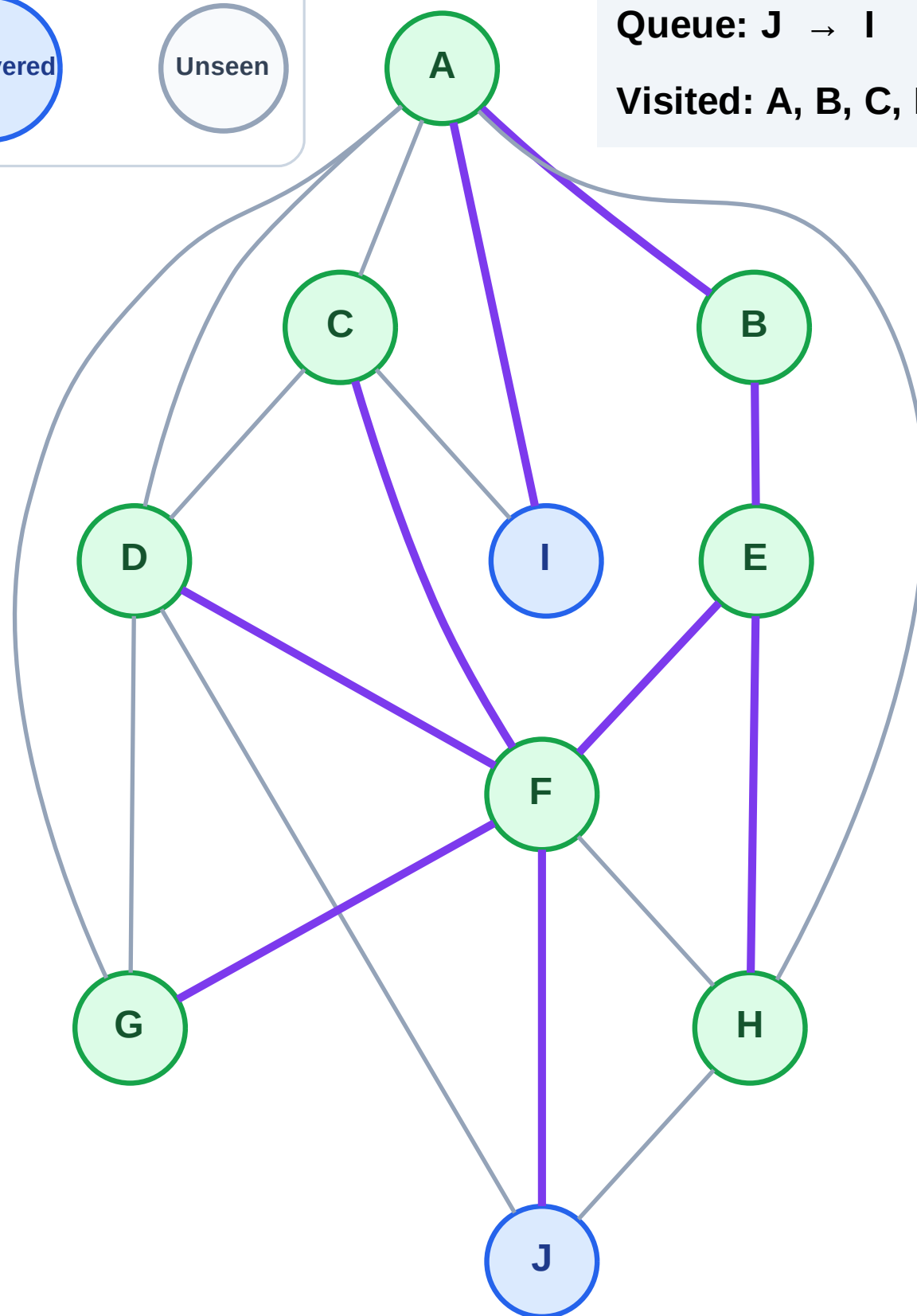
Step 17: No new neighbors from G

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

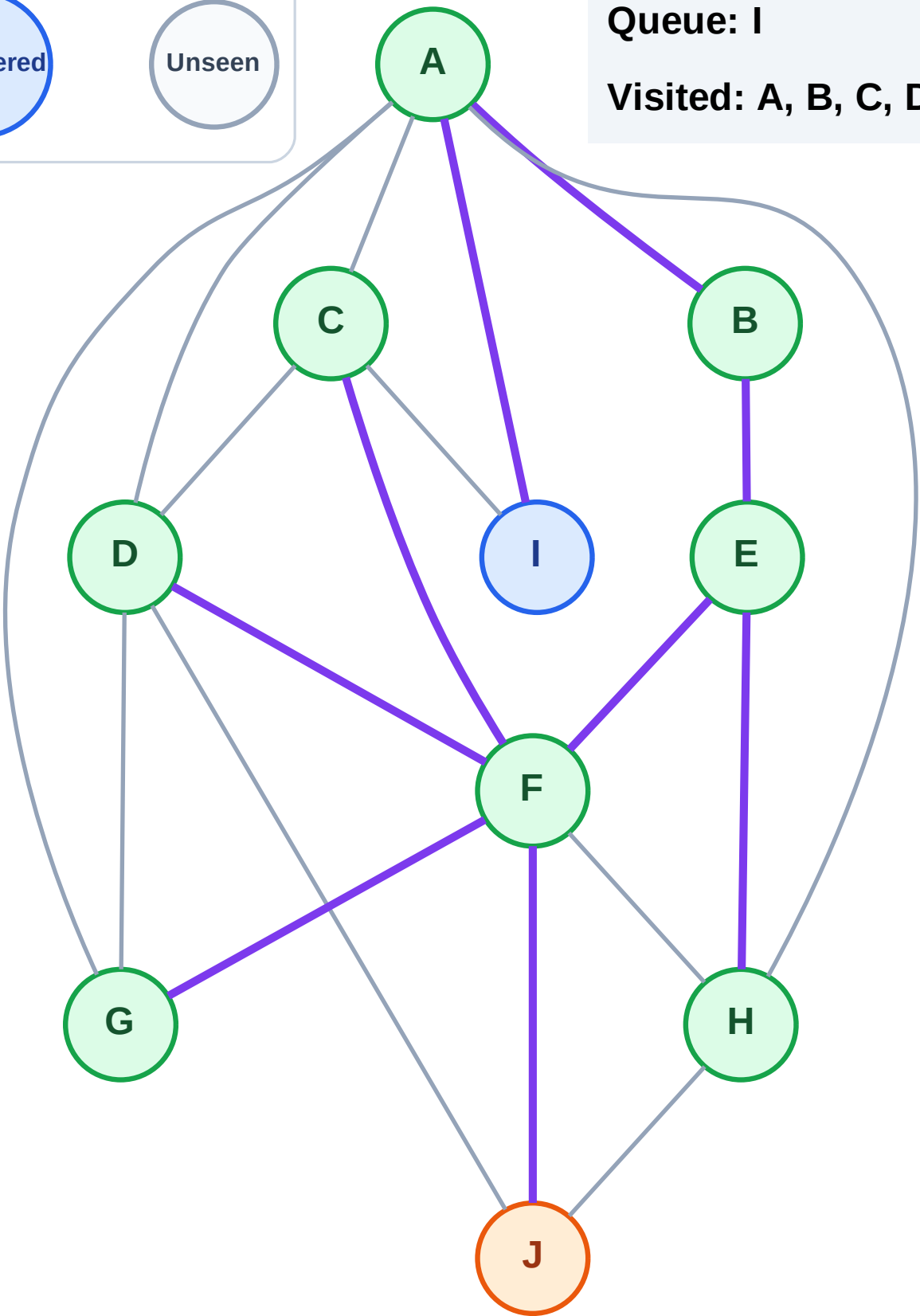
Processing

Discovered

Unseen

Queue: I

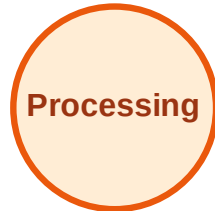
Visited: A, B, C, D, E, F, G, H



BFS Traversal

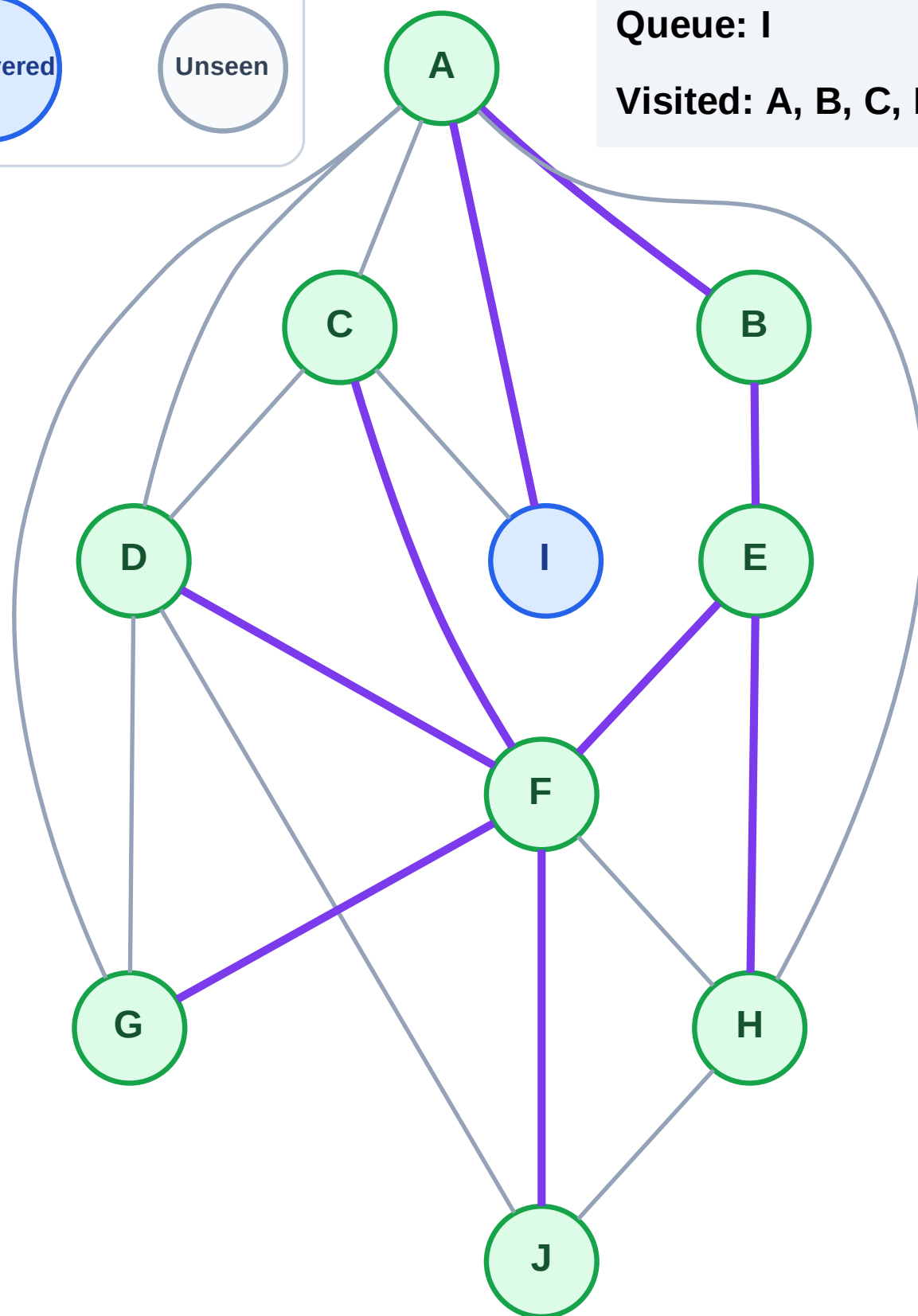
Step 19: No new neighbors from J

Legend



Queue: I

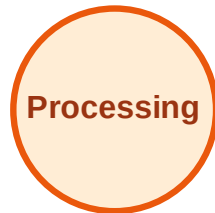
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

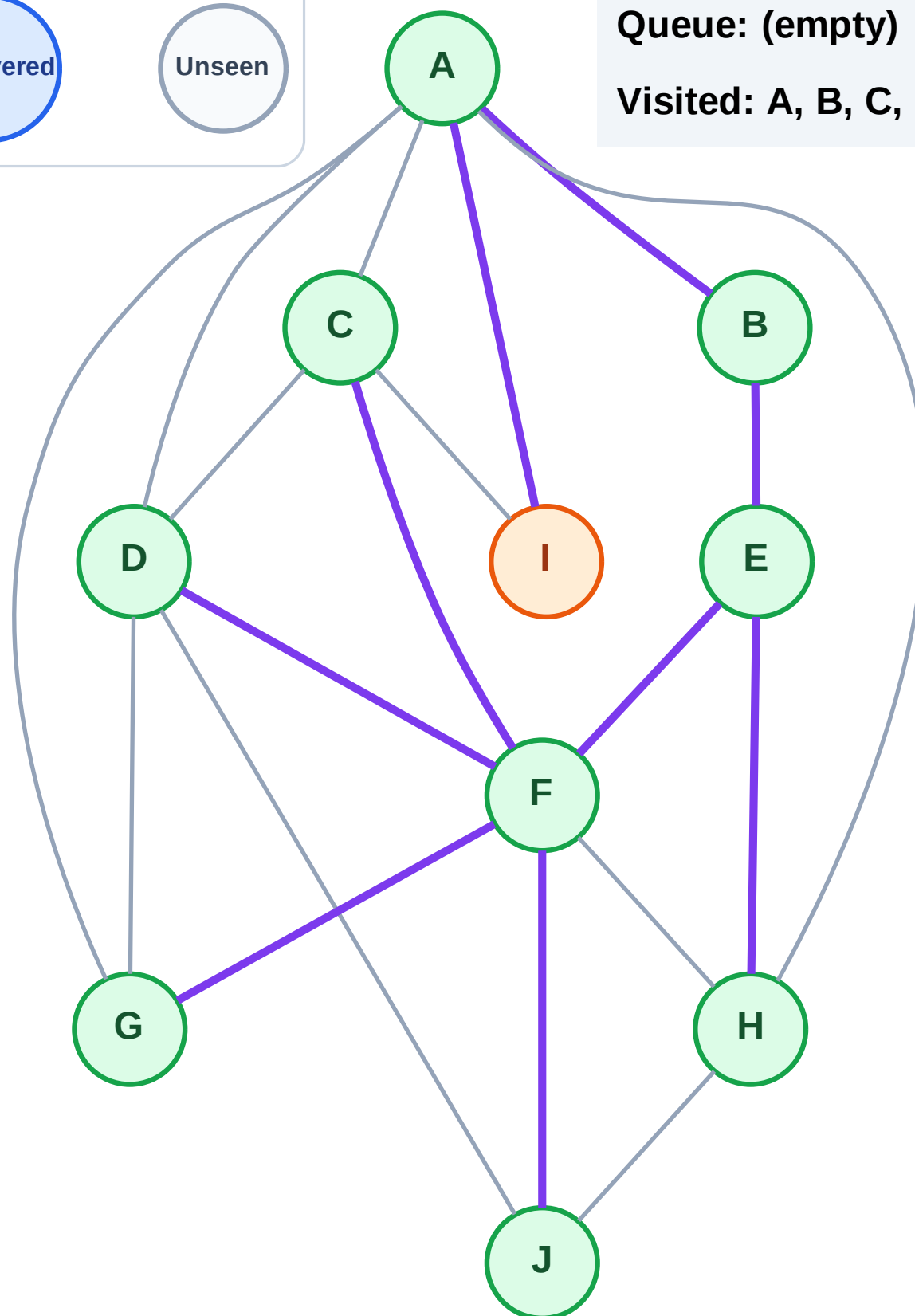
Step 20: Process node I

Legend



Queue: (empty)

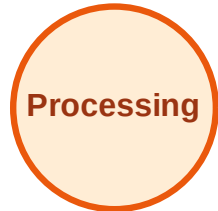
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

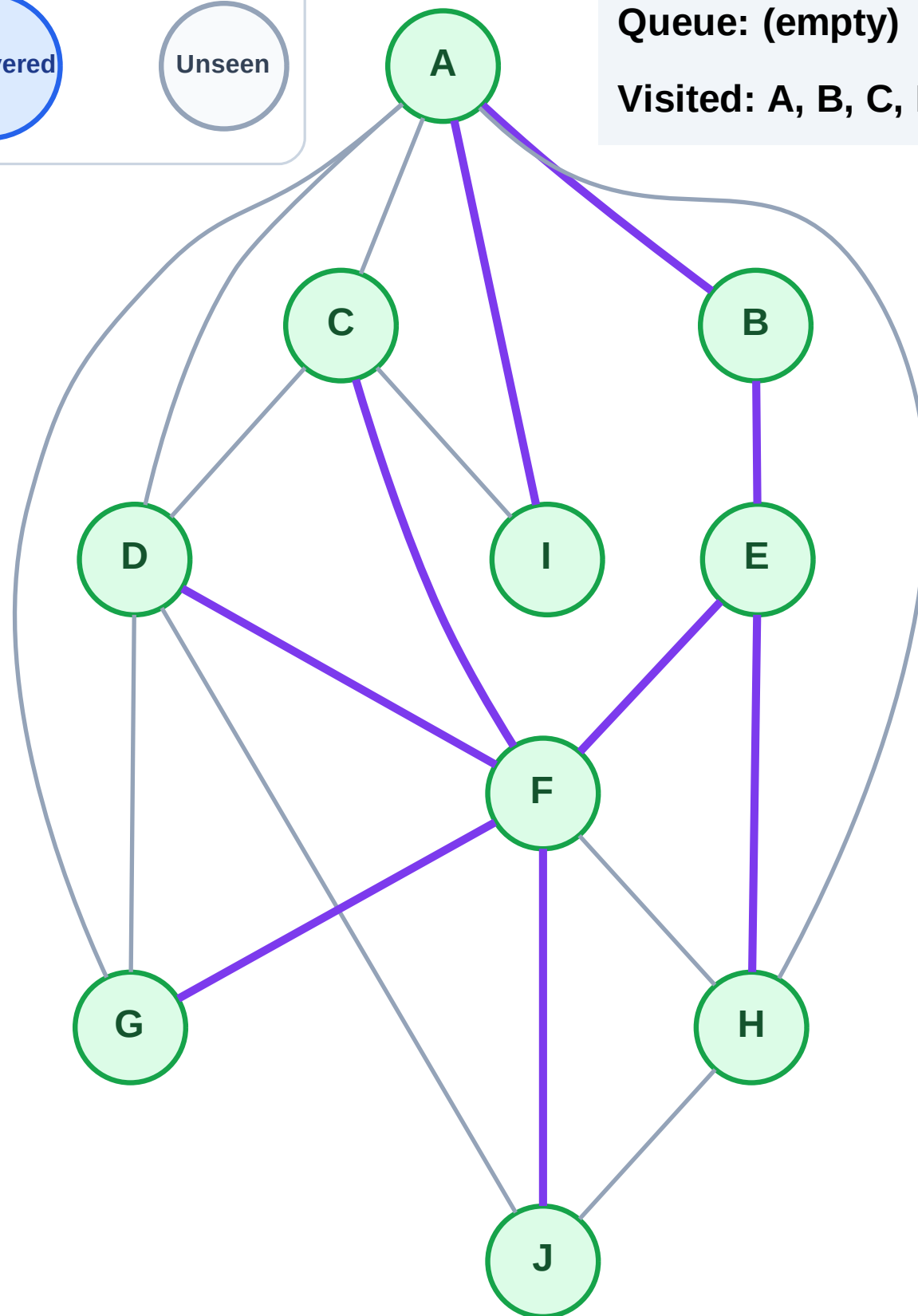
Step 21: No new neighbors from I

Legend



Queue: (empty)

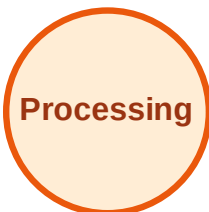
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

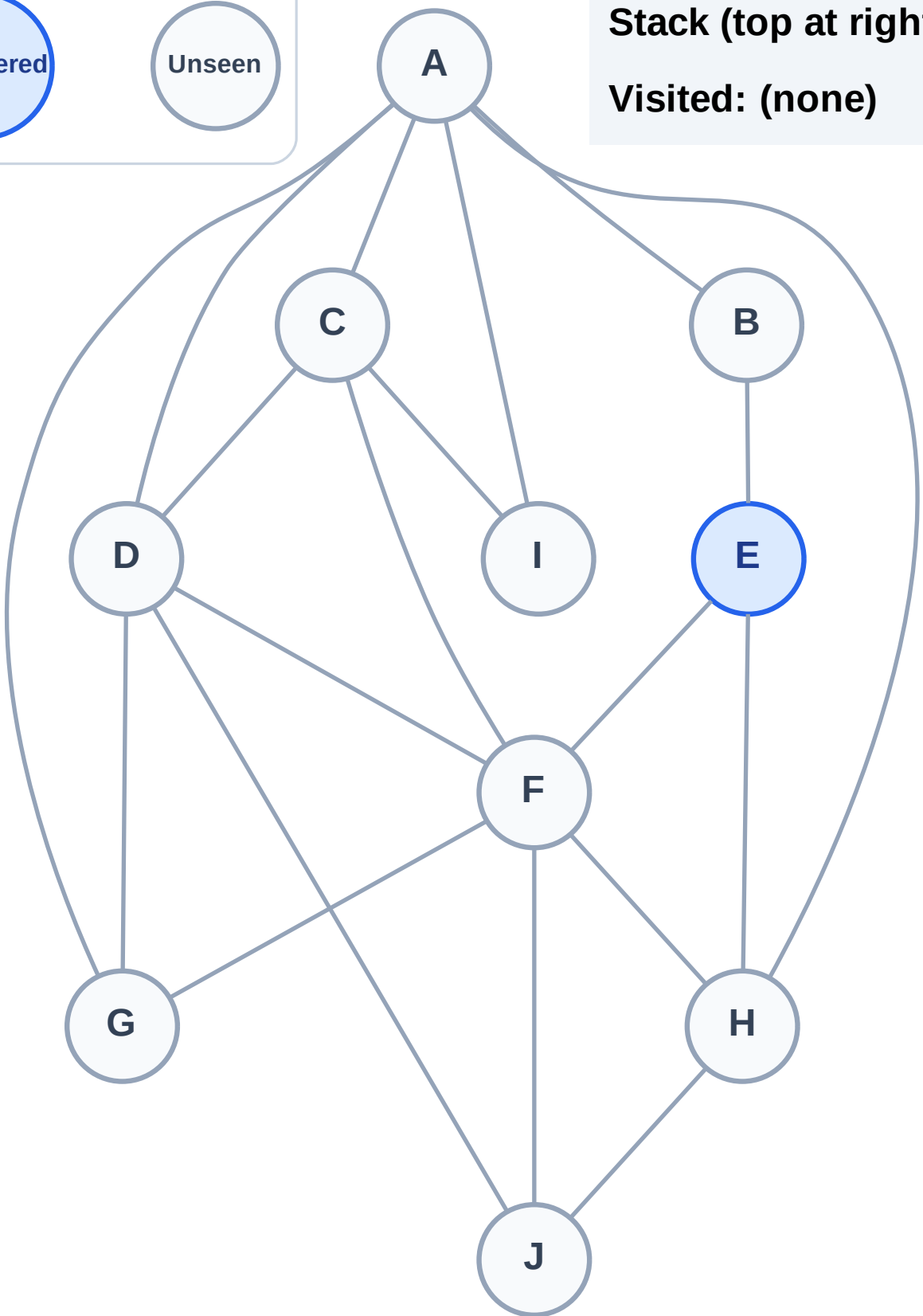
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

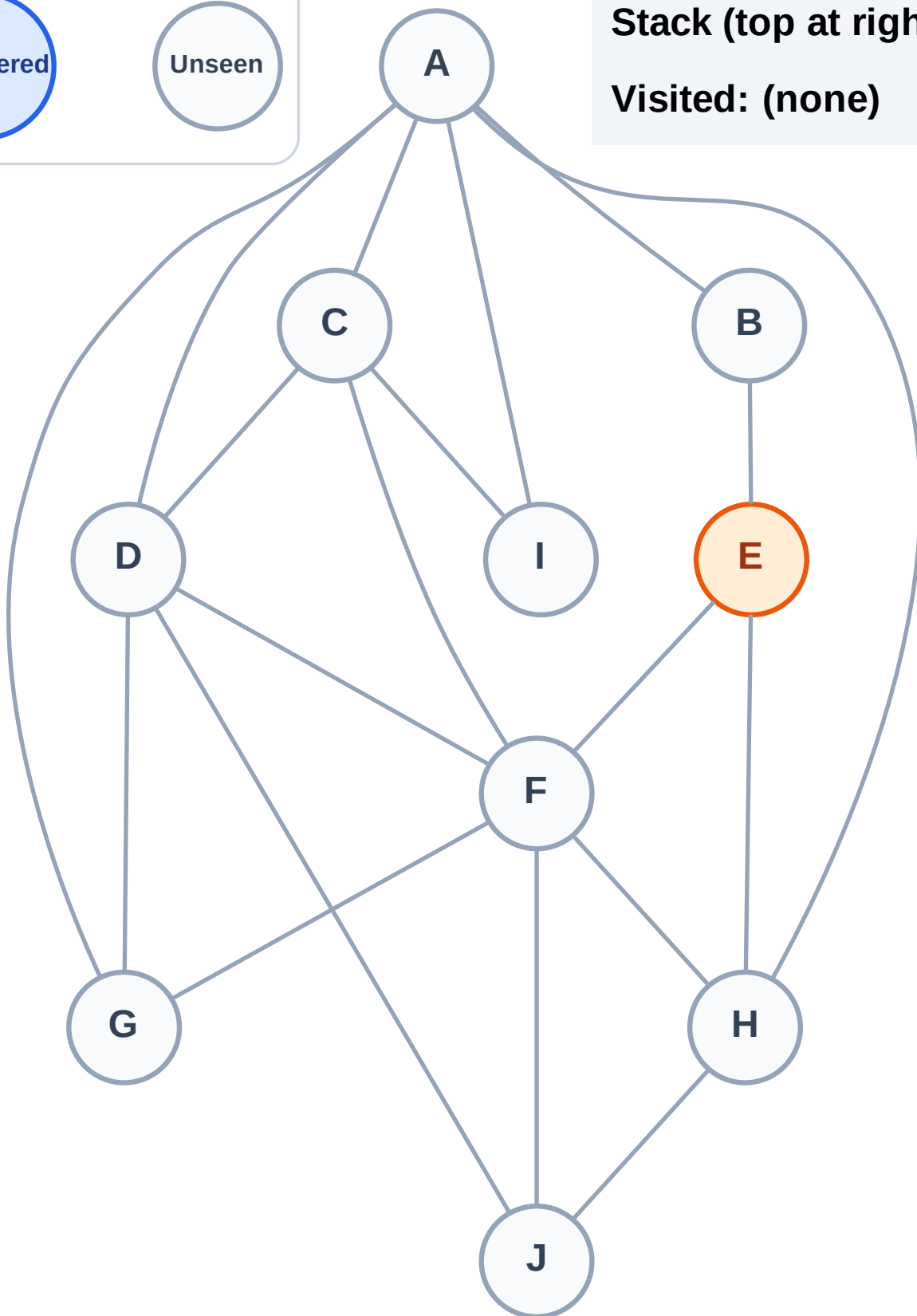
Processing

Discovered

Unseen

Stack (top at right): (empty)

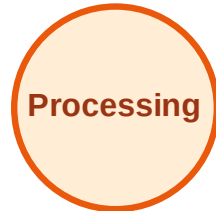
Visited: (none)



DFS Traversal

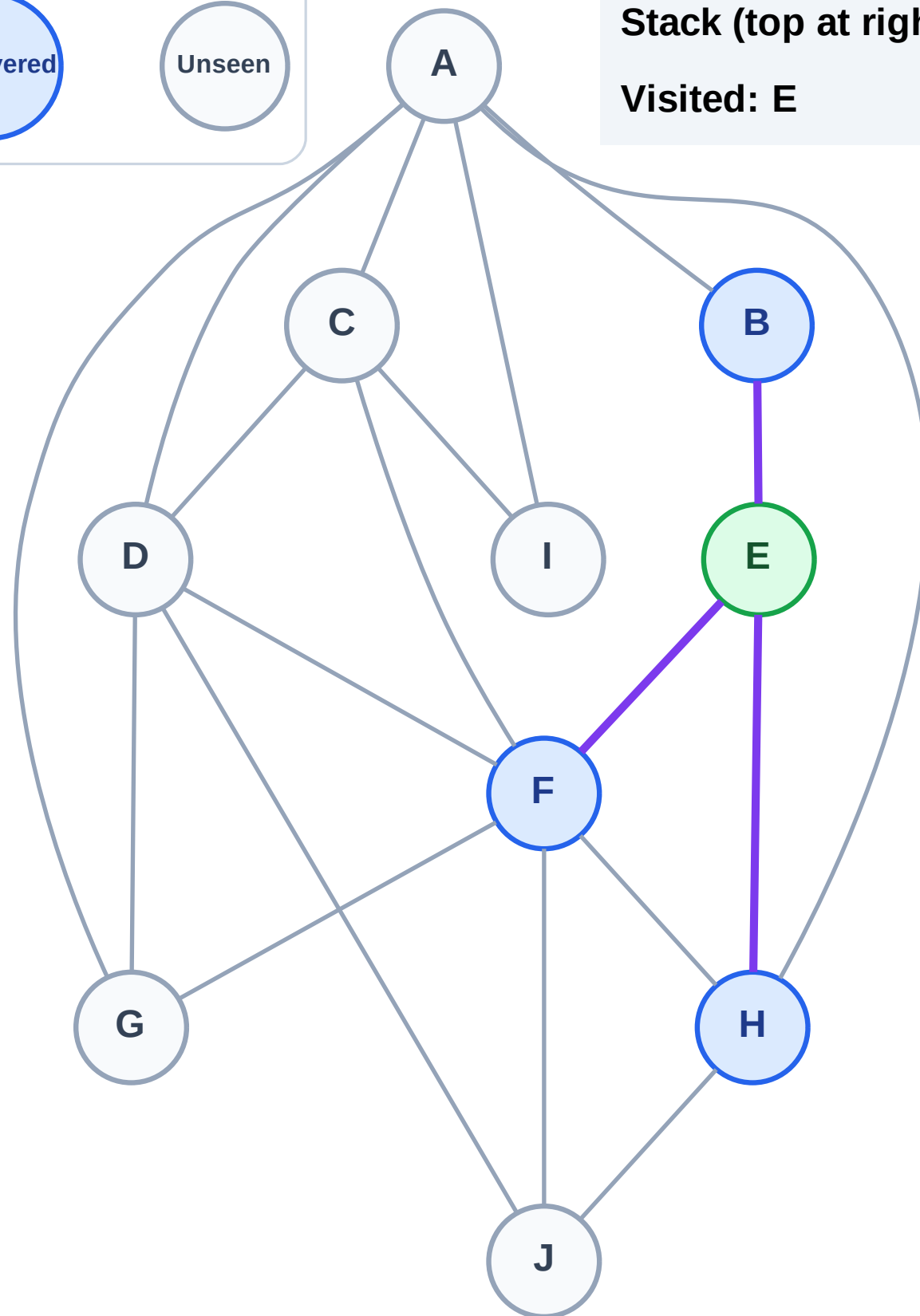
Step 3: Discover H, F, B from E

Legend



Stack (top at right): H | F | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

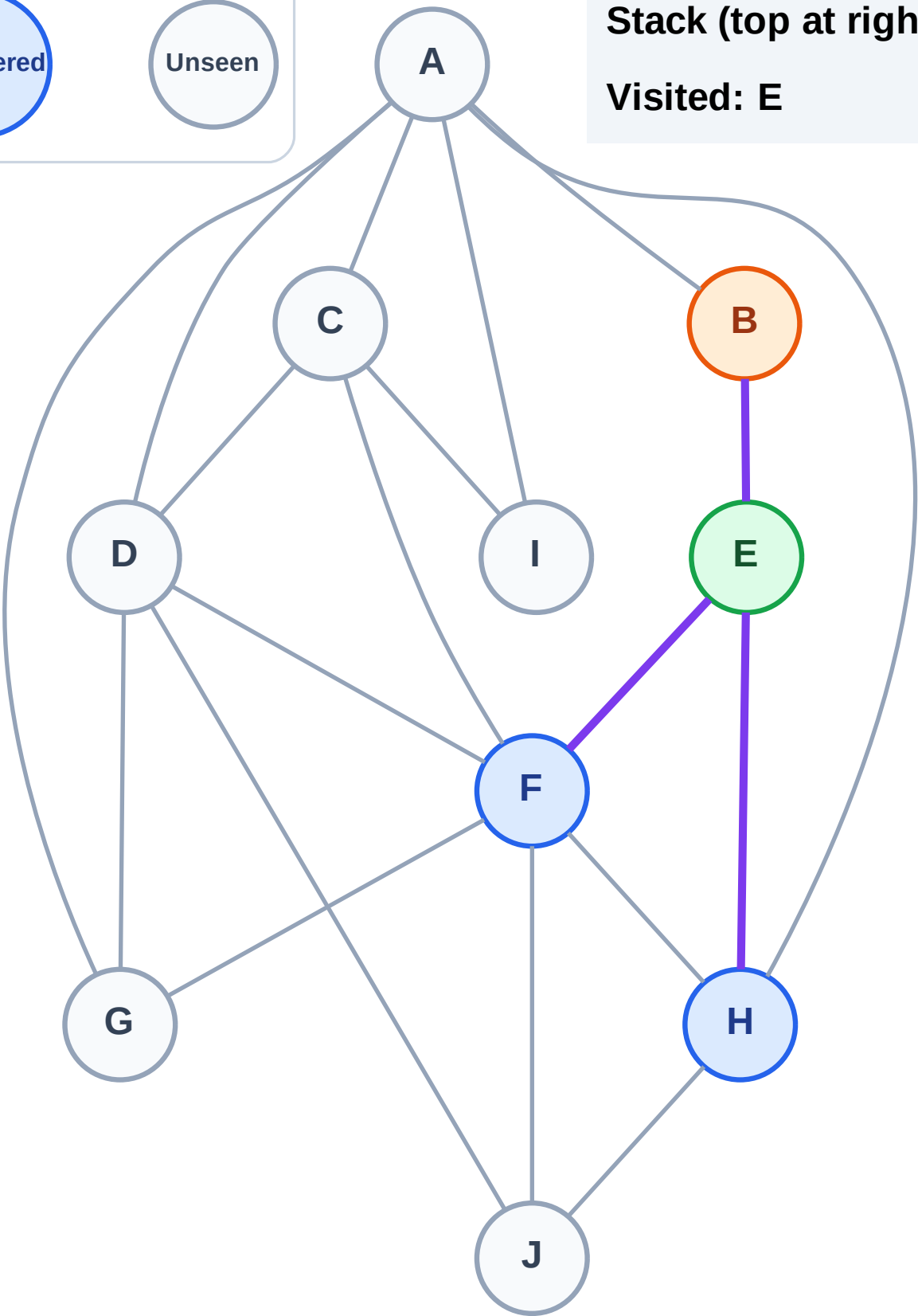
Complete

Processing

Discovered

Unseen

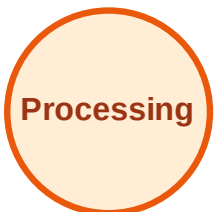
Stack (top at right): H | F
Visited: E



DFS Traversal

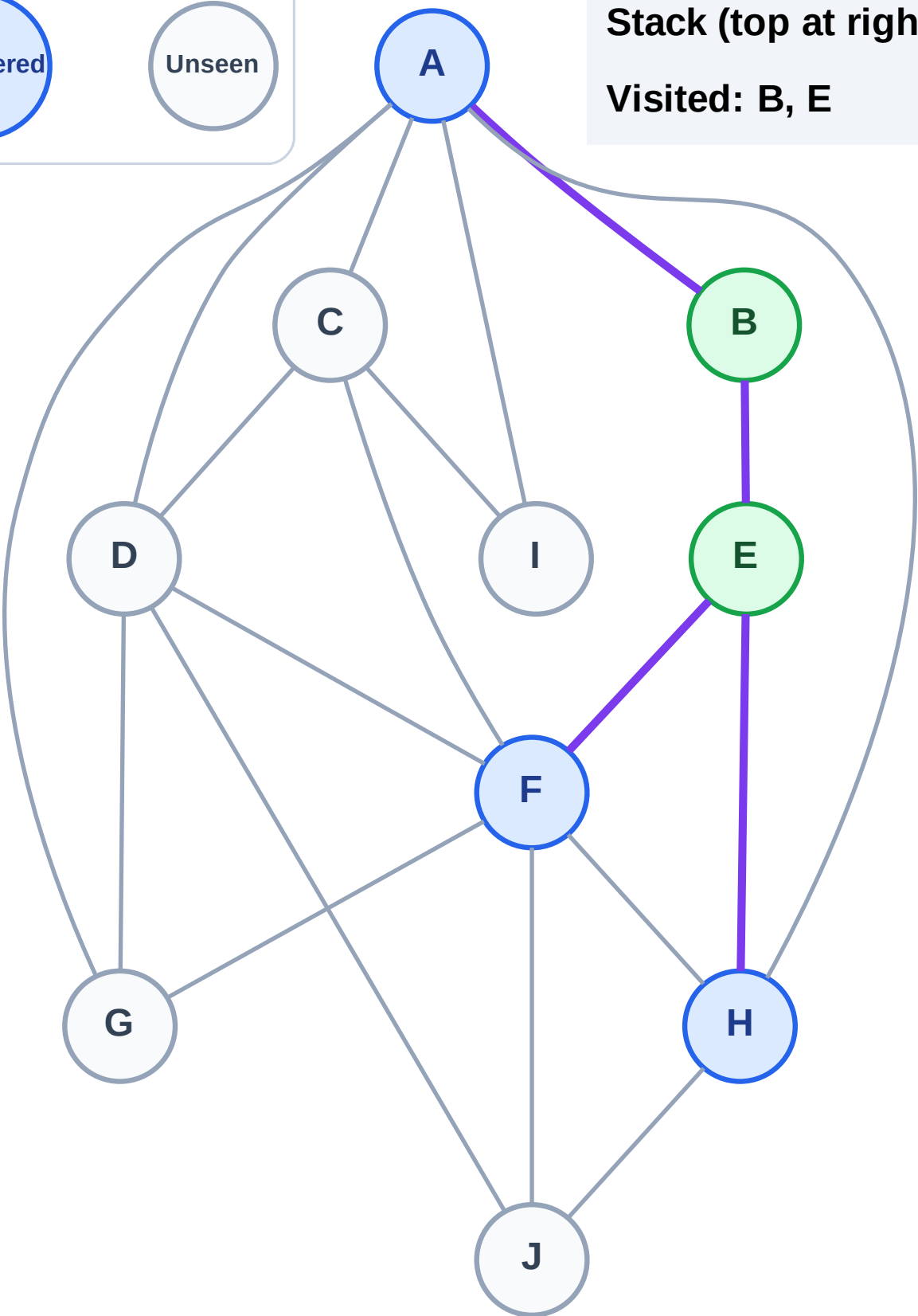
Step 5: Discover A from B

Legend



Stack (top at right): H | F | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

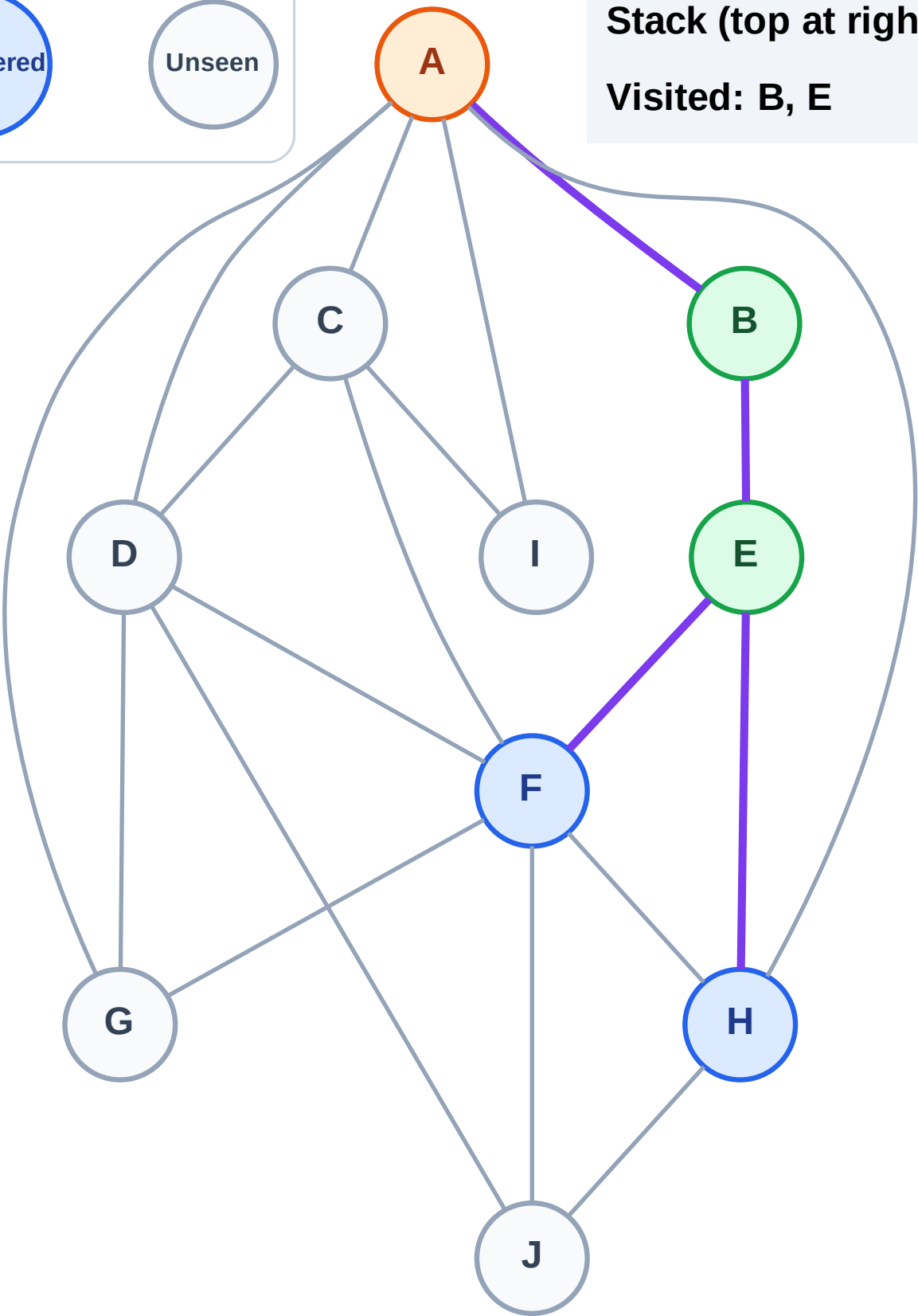
Complete

Processing

Discovered

Unseen

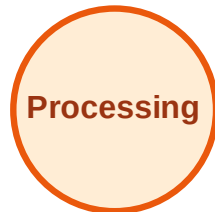
Stack (top at right): H | F
Visited: B, E



DFS Traversal

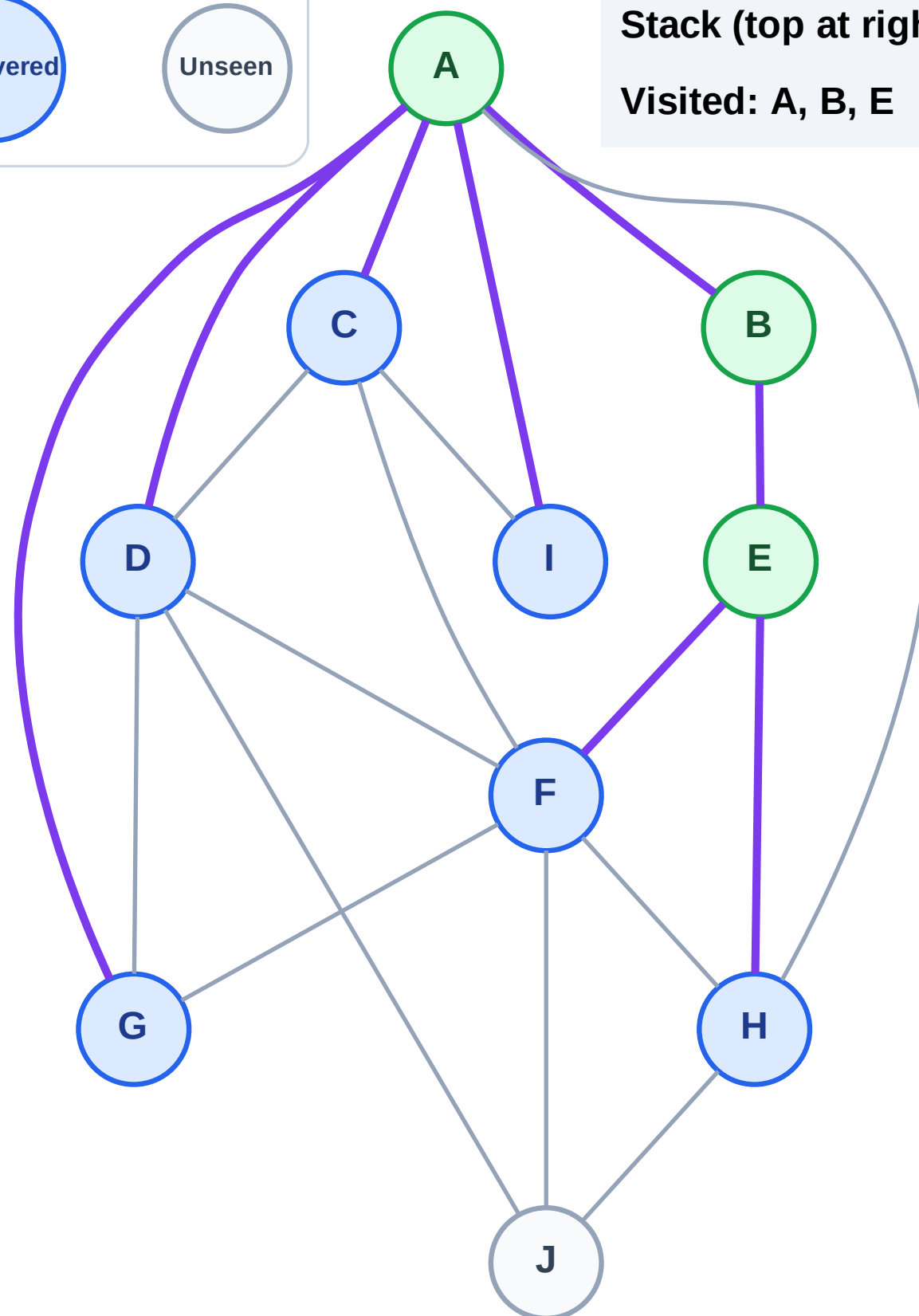
Step 7: Discover I, G, D, C from A

Legend



Stack (top at right): H | F | I | G | D | C

Visited: A, B, E



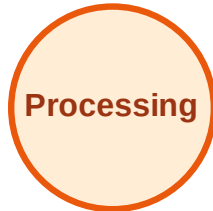
DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



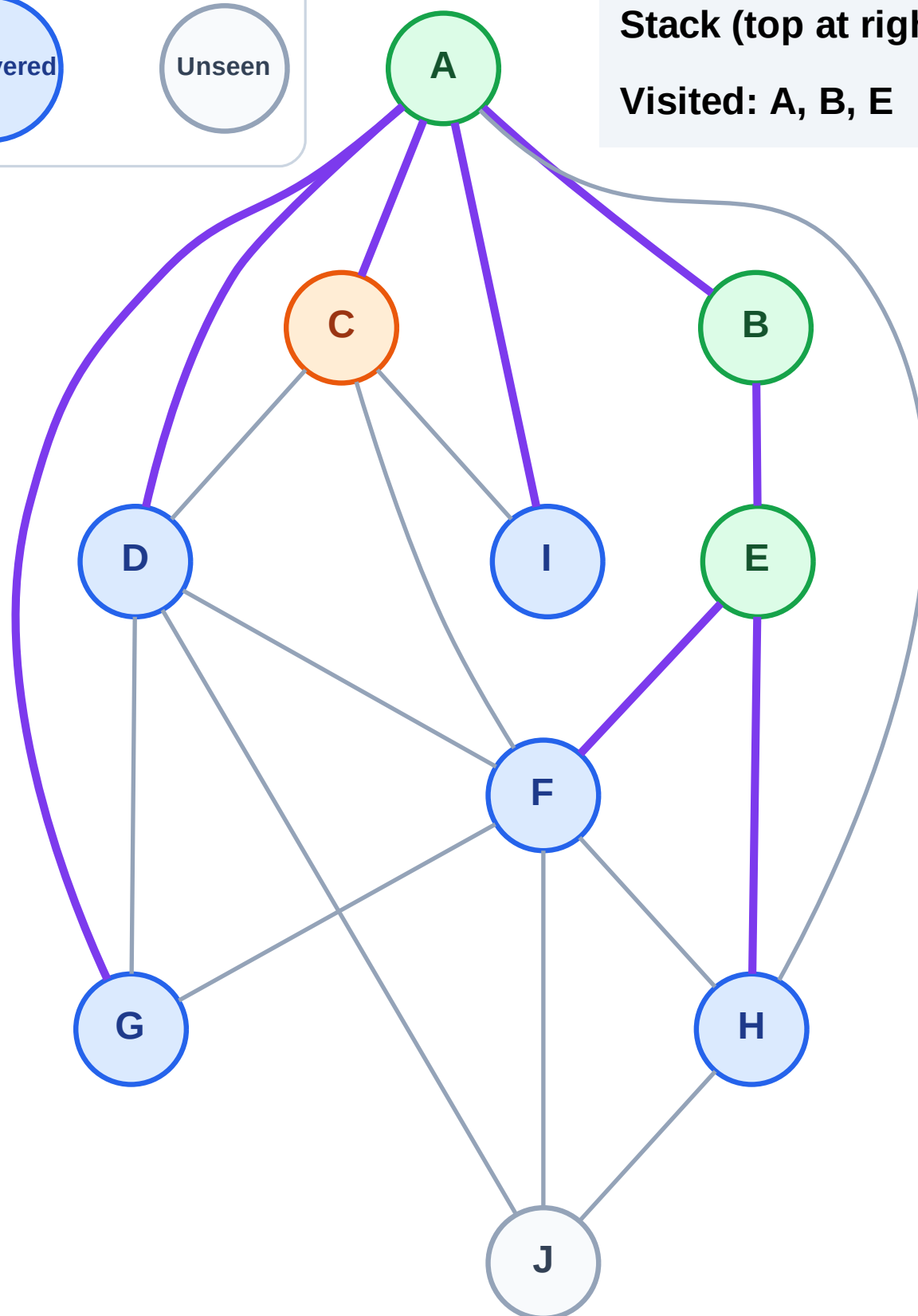
Discovered



Unseen

Stack (top at right): H | F | I | G | D

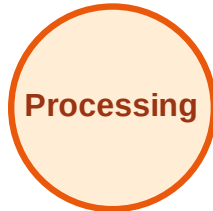
Visited: A, B, E



DFS Traversal

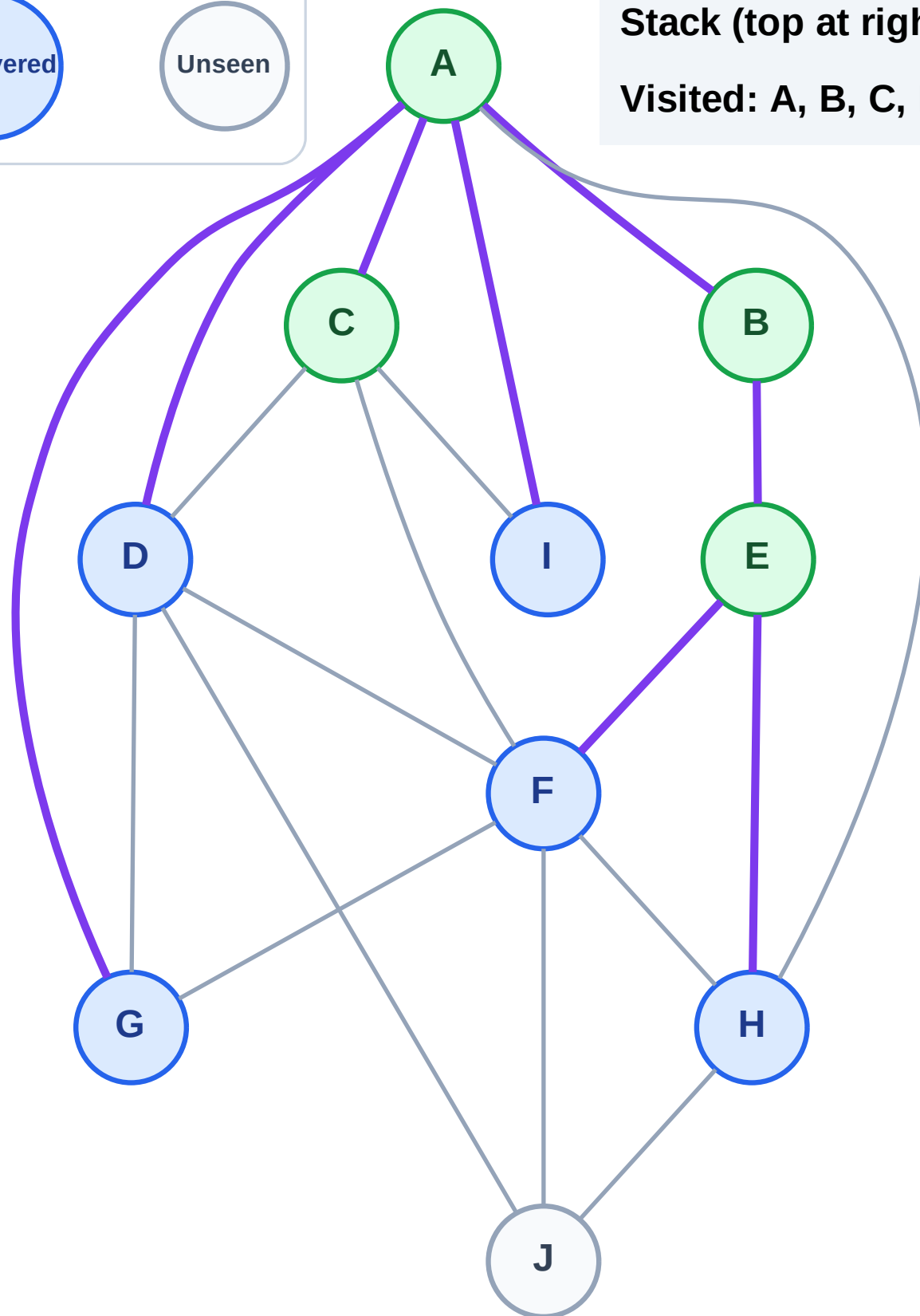
Step 9: No new neighbors from C

Legend



Stack (top at right): H | F | I | G | D

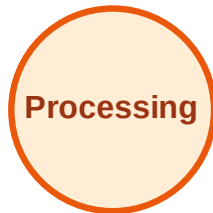
Visited: A, B, C, E



DFS Traversal

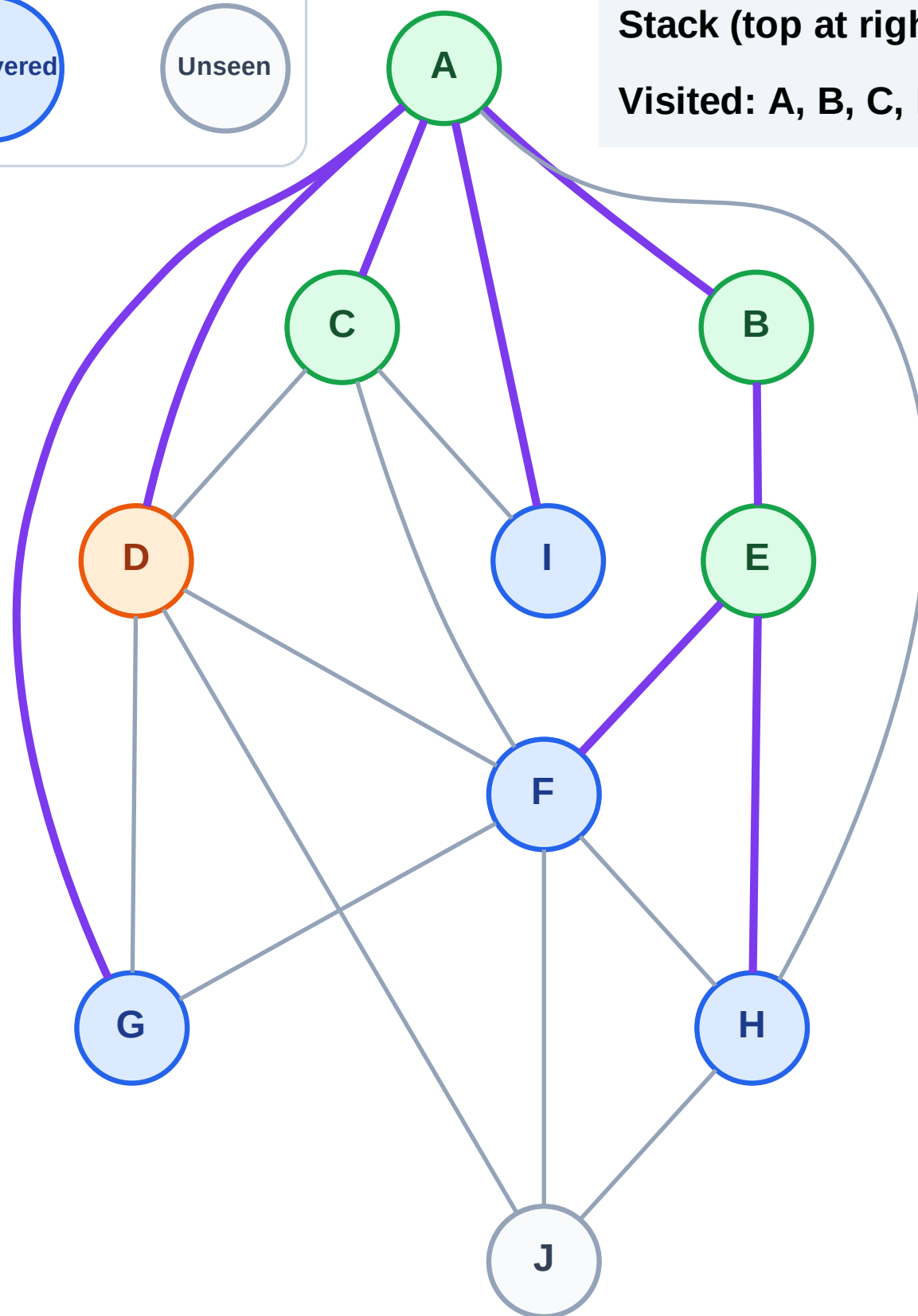
Step 10: Process node D

Legend



Stack (top at right): H | F | I | G

Visited: A, B, C, E



DFS Traversal

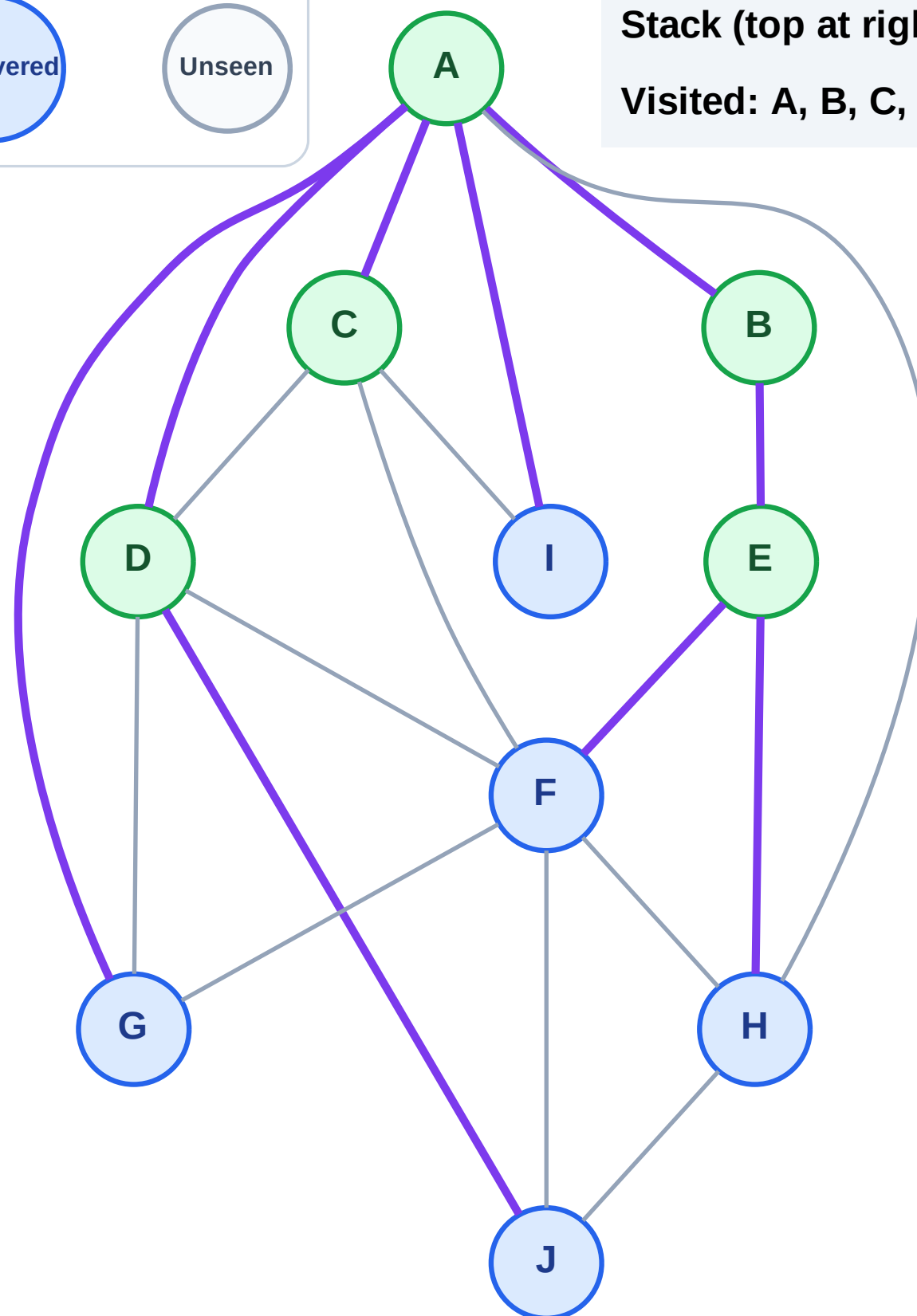
Step 11: Discover J from D

Legend



Stack (top at right): H | F | I | G | J

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node J

Legend

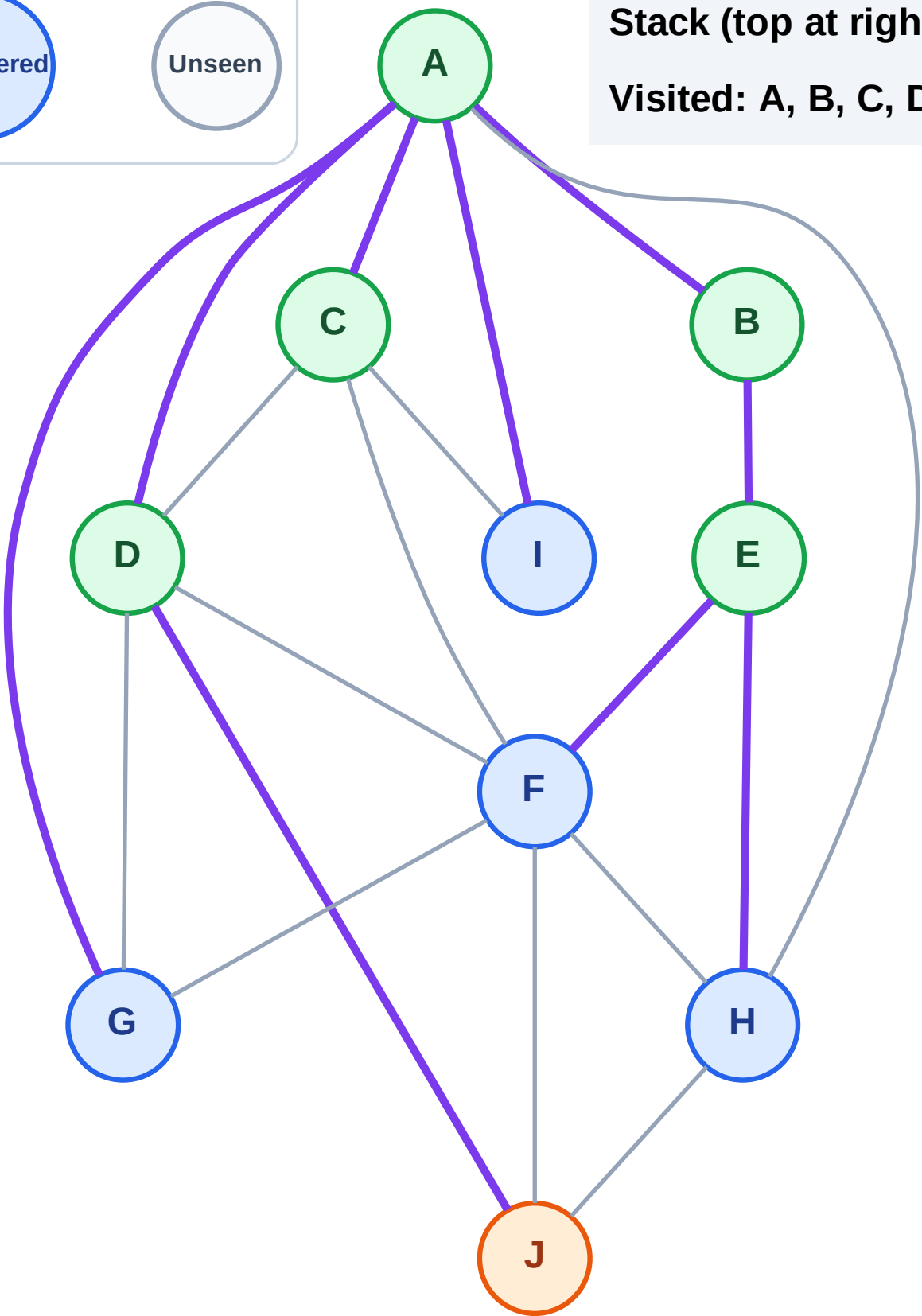
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | I | G
Visited: A, B, C, D, E



DFS Traversal

Step 13: No new neighbors from J

Legend

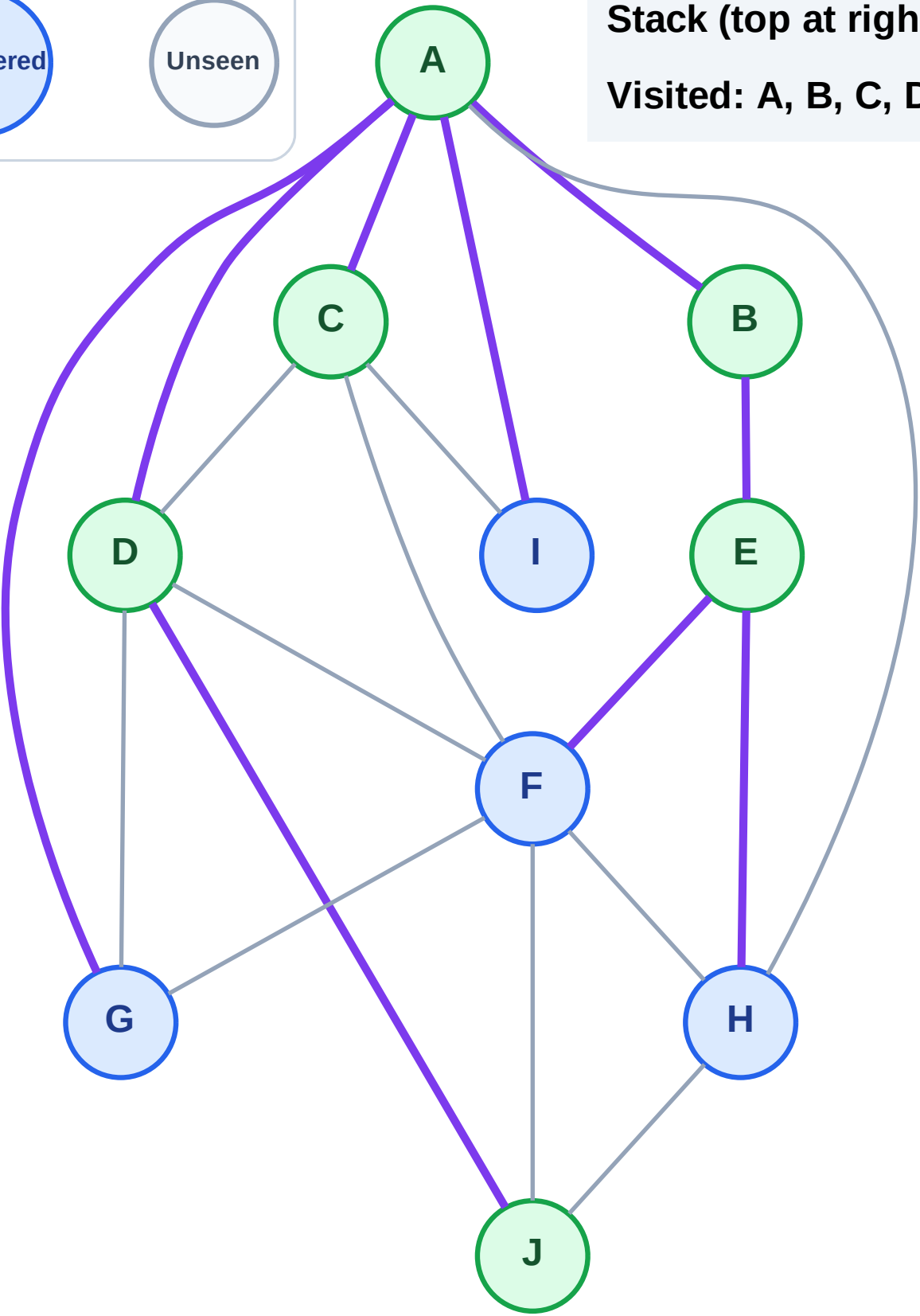
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | I | G
Visited: A, B, C, D, E, J



DFS Traversal

Step 14: Process node G

Legend

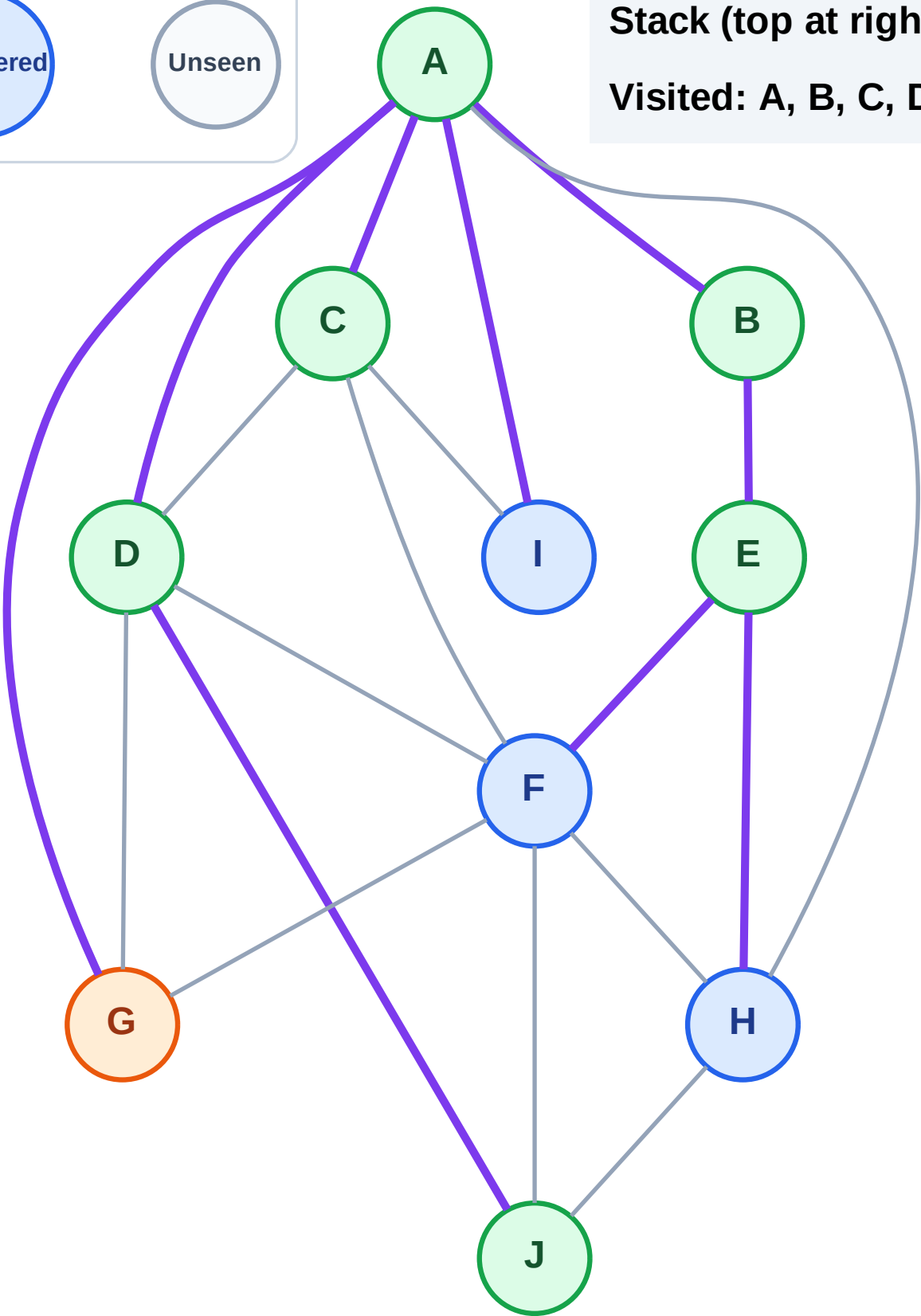
Complete

Processing

Discovered

Unseen

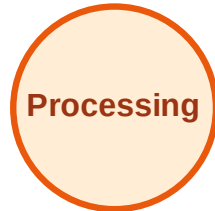
Stack (top at right): H | F | I
Visited: A, B, C, D, E, J



DFS Traversal

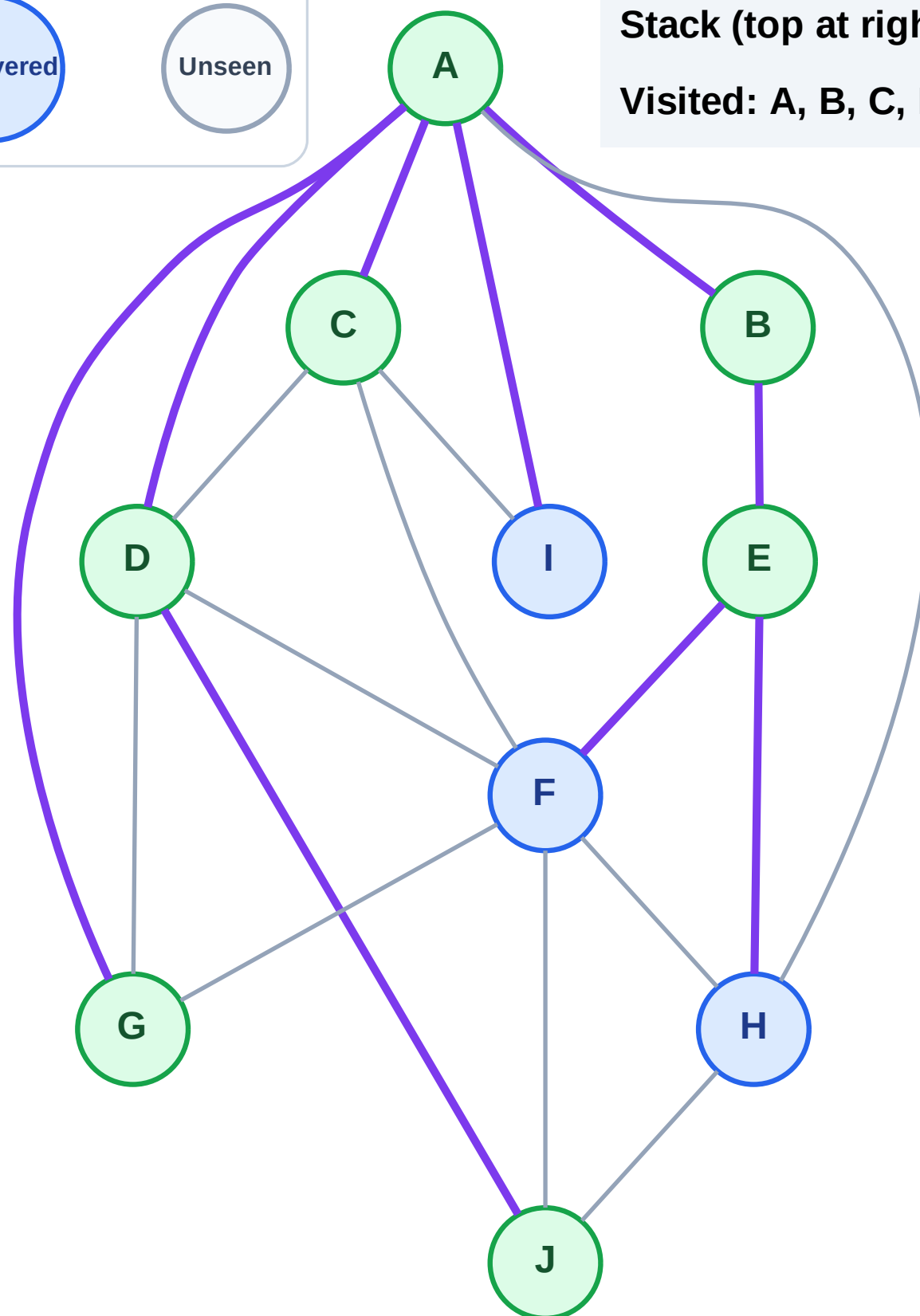
Step 15: No new neighbors from G

Legend



Stack (top at right): H | F | I

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node I

Legend

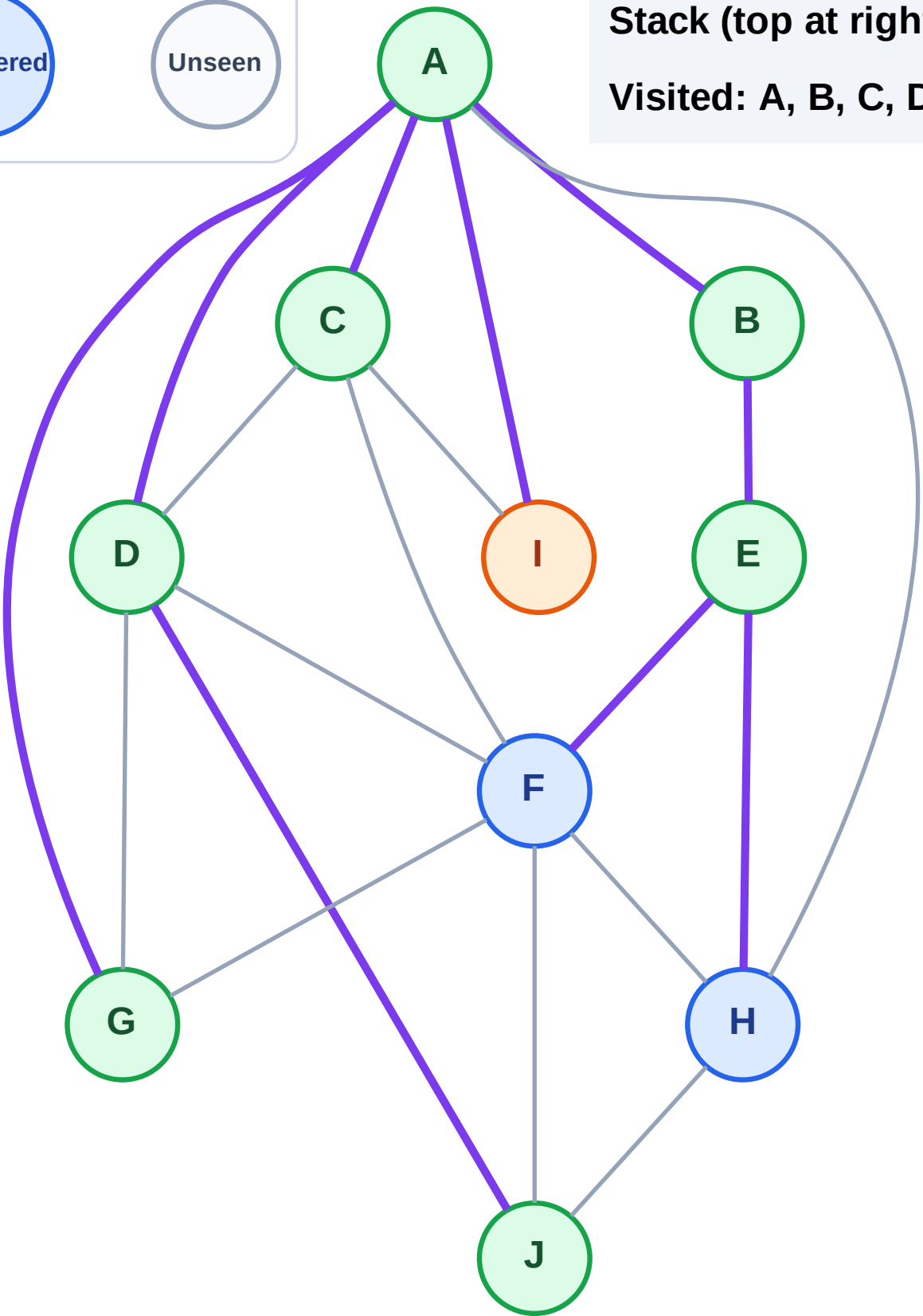
Complete

Processing

Discovered

Unseen

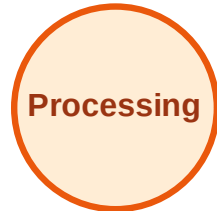
Stack (top at right): H | F
Visited: A, B, C, D, E, G, J



DFS Traversal

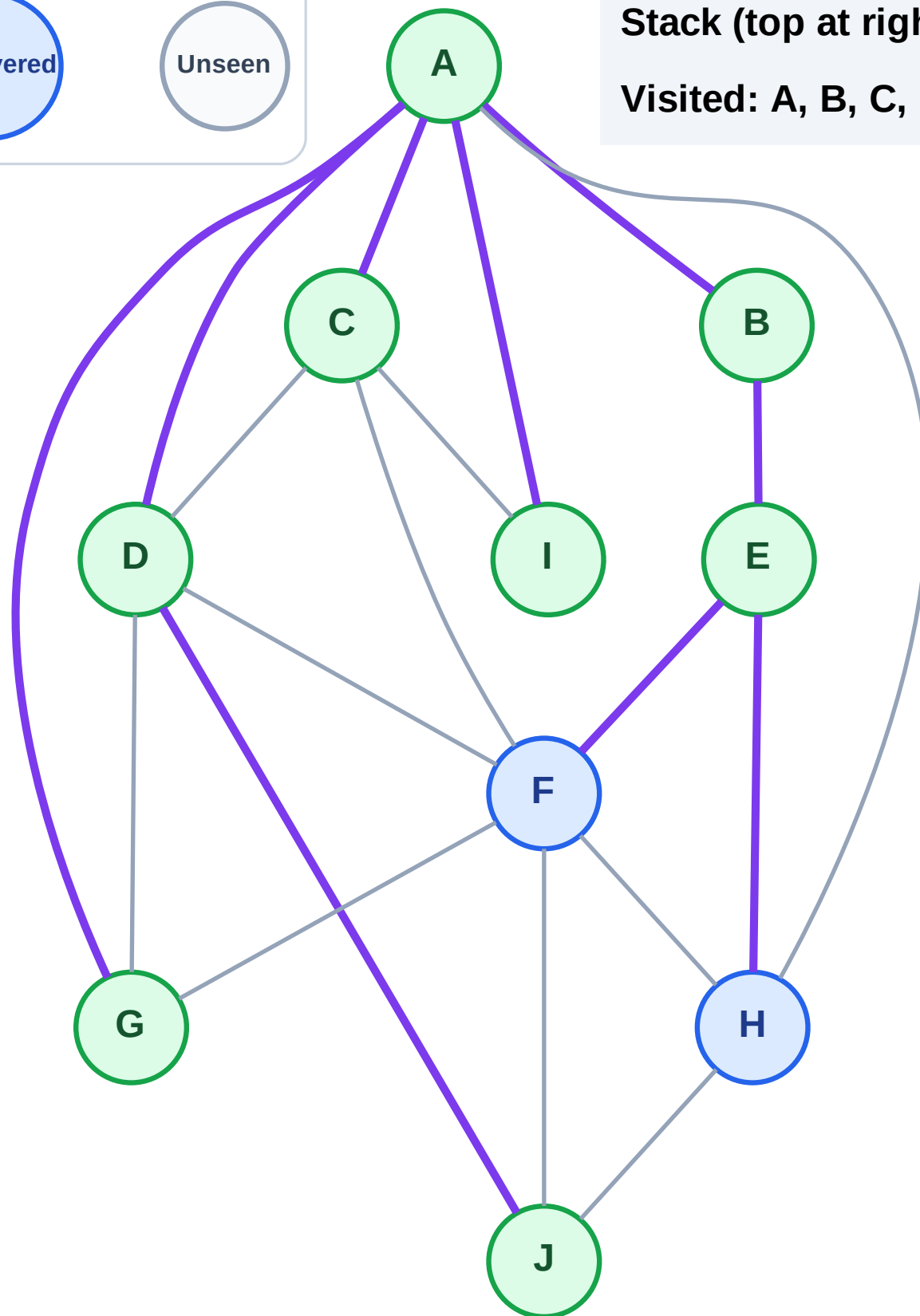
Step 17: No new neighbors from I

Legend



Stack (top at right): H | F

Visited: A, B, C, D, E, G, I, J



DFS Traversal

Step 18: Process node F

Legend

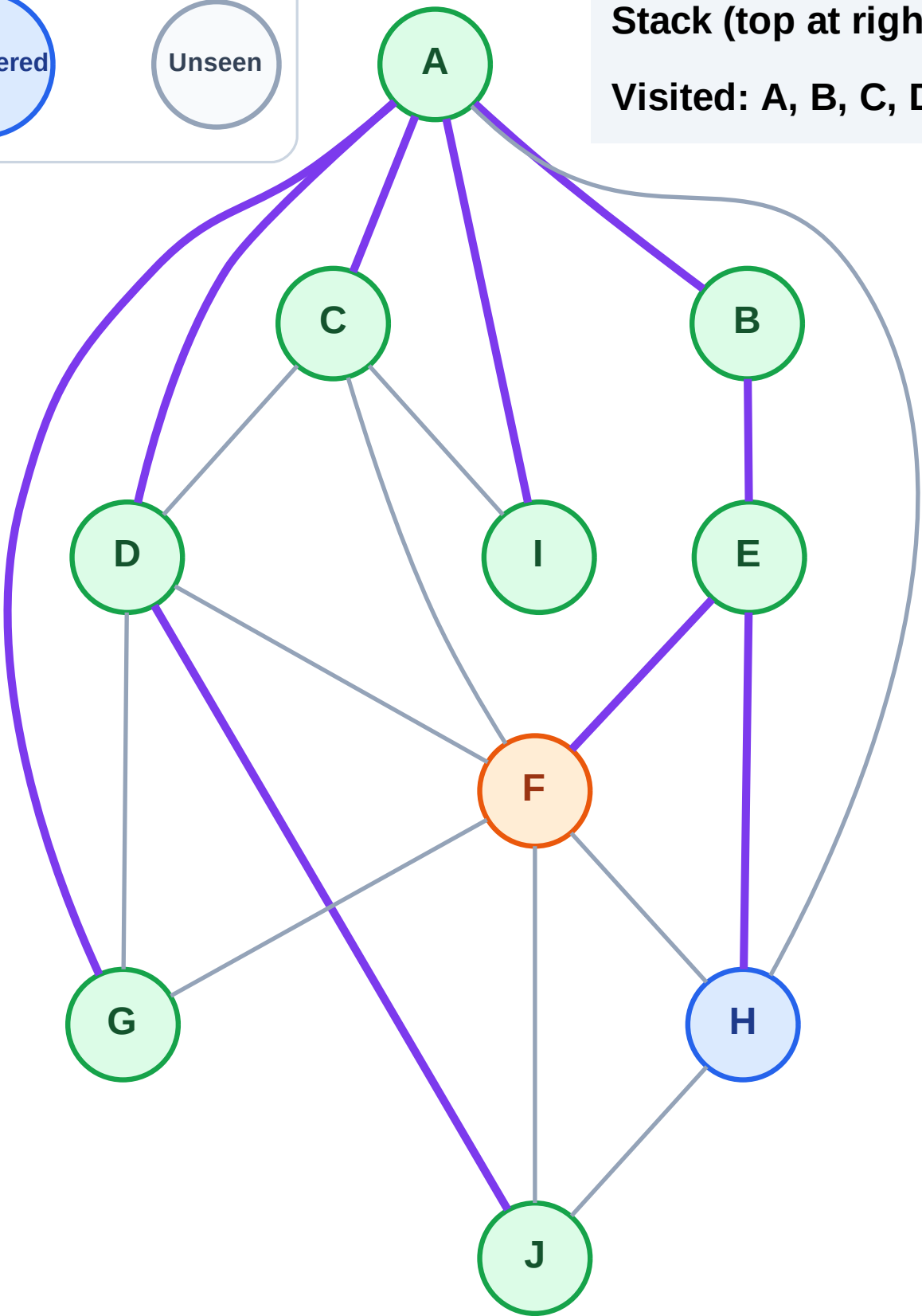
Complete

Processing

Discovered

Unseen

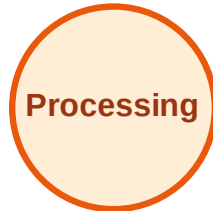
Stack (top at right): H
Visited: A, B, C, D, E, G, I, J



DFS Traversal

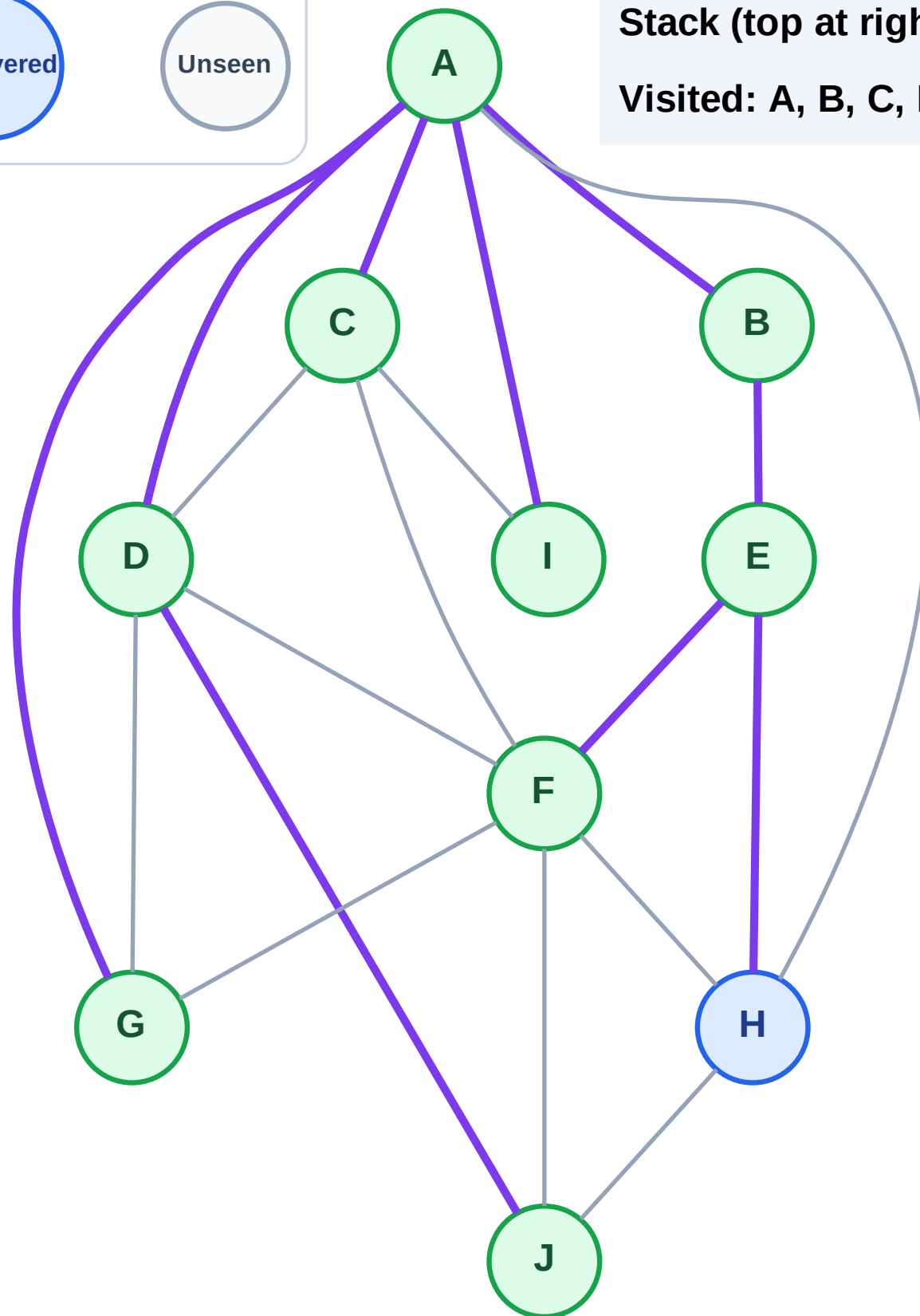
Step 19: No new neighbors from F

Legend



Stack (top at right): H

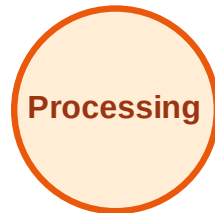
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

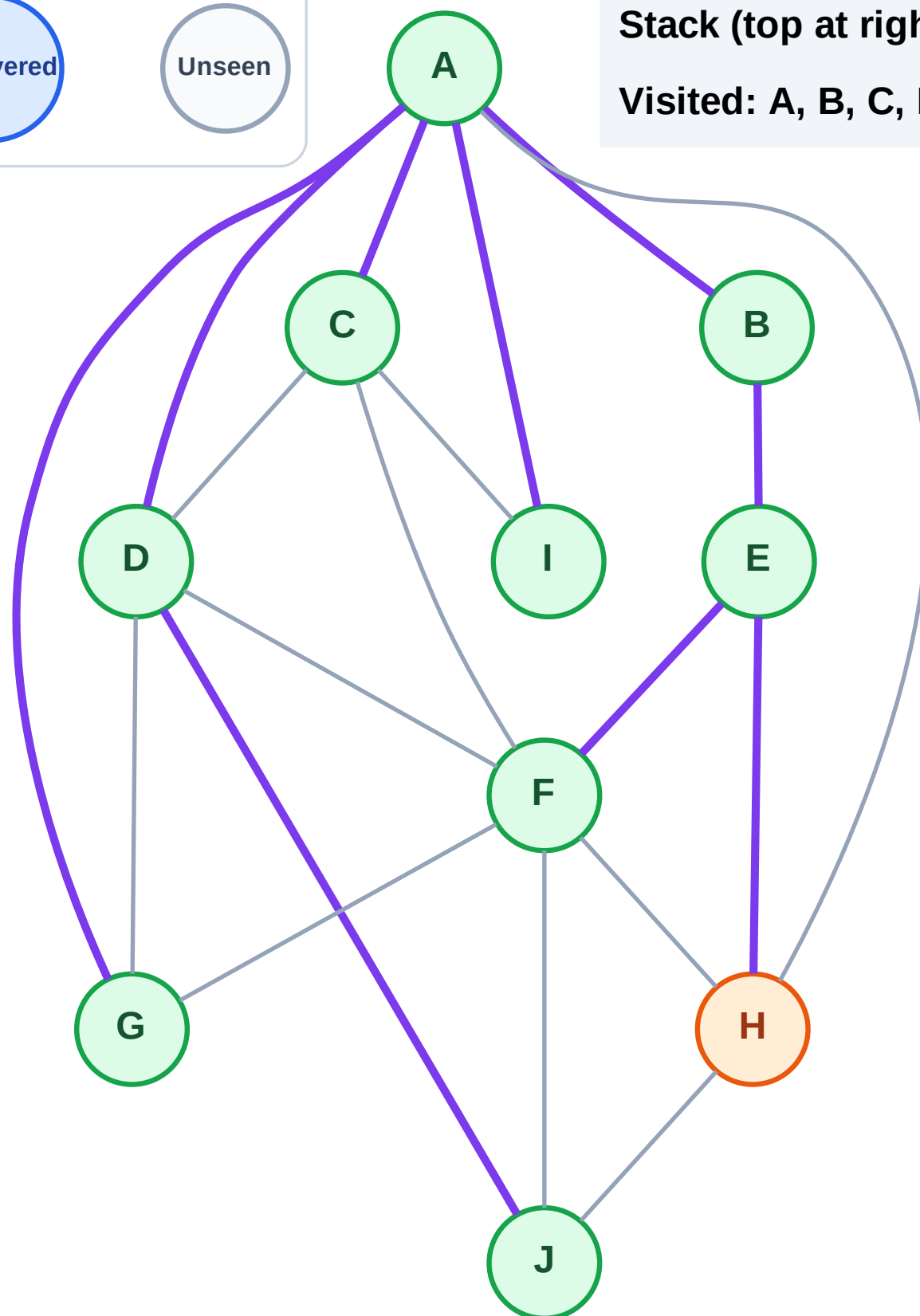
Step 20: Process node H

Legend



Stack (top at right): (empty)

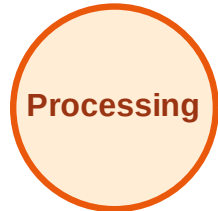
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

